



Review article

Decoding synaptic imbalance in neurodegenerative diseases: From pathological analysis to targeted intervention

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ABSTRACT

Synapses serve as the central functional components mediating information transmission, integration, and storage within the central nervous system (CNS). Their functionality depends on the synergistic interplay of the presynaptic membrane, synaptic cleft, and postsynaptic membrane—three structures that collectively sustain neurotransmitter secretion, postsynaptic signaling, and synaptic plasticity. Of note, synaptic impairment represents an early, shared pathological hallmark across aging and age-related neurodegenerative disorders like Alzheimer's disease (AD), Parkinson's disease (PD), Huntington's disease (HD), and Amyotrophic lateral sclerosis (ALS). This damage emerges prior to the demise of neuronal cell bodies and the manifestation of clinical symptoms, with its location and severity directly shaping disease phenotypes. Importantly, such synaptic dysfunction is closely linked to the pathological proteins specific to each disorder. This review comprehensively synthesizes current research advances regarding synaptic impairment in age-related neurodegenerative diseases. It elaborates on the location-specific pathological features of synaptic damage in AD, PD, HD, and ALS, with particular emphasis on their associations with disease-related pathological markers. Additionally, the work unpacks five core mechanisms driving synaptic dysfunction: the toxic effects of pathological proteins, dysregulated synaptic protein homeostasis, excitotoxicity coupled with disrupted calcium balance, oxidative stress and inflammatory responses, and deficiencies in neurotrophic factors. Ultimately, it summarizes potential biomarkers and targeted intervention strategies, aiming to provide a systematic reference for mechanistic investigations, early diagnosis, and clinical management of neurodegenerative diseases.

1. Introduction

Synapses are specialized junctions that link neurons to one another or to effector cells, functioning as key components in the transmission, integration, and storage of information within the central nervous system (CNS) (Xiong et al., 2024). Under normal physiological circumstances, typical chemical synapses achieve neurotransmitter release, postsynaptic signal transduction, and regulation of synaptic plasticity via the coordinated interplay of the presynaptic membrane, synaptic cleft, and postsynaptic membrane. Of these processes, synaptic plasticity serves as the molecular underpinning of learning, memory, and neural development (Rizo, 2022).

In neurodegenerative disorders like Alzheimer's disease (AD), Parkinson's disease (PD), Huntington's disease (HD), and Amyotrophic lateral sclerosis (ALS), synaptic impairment emerges as an early and

prevalent pathological hallmark (Wilson et al., 2023). Synaptic impairment arises prior to both the death of neuronal somas and the manifestation of clinical symptoms, with its location and severity being closely linked to disease phenotypes. For instance, early synaptic loss in the hippocampus and entorhinal cortex in AD can lead to memory impairment (Ribarić, 2023); damage to dopaminergic synapses in the substantia nigra and striatum in PD causes bradykinesia (Dickson, 2018); preferential damage and destruction of synapses in the striatum in HD induce choreiform movements (Reiner and Deng, 2018); and damage to neuromuscular junctions (NMJs) in ALS directly results in muscle weakness (Valdez et al., 2012). Current studies have confirmed that such damage is closely associated with disease-specific pathological proteins (Bezprozvanny and Hiesinger, 2013), including A β oligomers (A β O) and tau tangles in AD (Watamura et al., 2025), α -synuclein (α -syn) aggregates in PD (Calabresi et al., 2023), and mutant huntingtin

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(mHTT) in HD (Gu et al., 2025). Even so, key questions remain unanswered and call for systematic collation and in-depth exploration: how pathological proteins trigger synaptic structural impairment and functional anomalies via diverse pathways, what shared mechanisms and distinct characteristics underlie synaptic damage across different diseases, and how to develop early diagnostic biomarkers and targeted protective strategies based on these mechanisms.

Aging is a major risk factor for neurodegenerative diseases. Considering the central role of synaptic damage in the pathological progression of age-related neurodegenerative diseases, this review centers on the four aforementioned neurodegenerative diseases and systematically collates the site-specificity of synaptic impairment, its structural and functional alterations, and correlations with pathological markers. It further delves into five core damaging mechanisms, including the toxic impacts of pathological proteins, disruptions to synaptic protein homeostasis, excitotoxicity, oxidative stress and inflammatory responses, and neurotrophic factor insufficiency. Building on this foundation, the review outlines the biomarkers detectable in cerebrospinal fluid (CSF), blood, and neuroimaging for the early diagnosis of synaptic damage. Finally, it puts forward targeted therapeutic approaches, including inhibiting the production and aggregation of pathological proteins, regulating synaptic homeostasis, and mitigating excitotoxicity, aiming to offer a comprehensive reference for mechanistic research, early diagnosis, and clinical intervention in neurodegenerative diseases.

2. Pathological characteristics of synaptic damage in neurodegenerative diseases

This section systematically reviews the site-specific pathological characteristics of synaptic damage in age-related neurodegenerative diseases, and focuses on analysing and identifying synaptic impairment as a unifying early pathological hallmark of neurodegenerative diseases. This site-specific analysis not only clarifies the pathogenic basis of disease phenotypes but also provides a framework for understanding disease progression.

2.1. Synaptic damage in AD

Alzheimer's disease (AD) is the most prevalent neurodegenerative disorder, defined by gradual memory deterioration and cognitive impairment as its primary clinical manifestations (Schirge et al., 2025; Selma-Gonzalez et al., 2025; Sun et al., 2024b). Synaptic damage in AD exhibits marked site-specificity and temporal dependency, and these two features are closely linked to the condition's progression and clinical manifestations (Cheng et al., 2025). Abnormal phosphorylation of tau protein is the key driver of synaptic damage in AD (John and Reddy, 2021). Through synaptic morphological analysis, Dr. Reddy's team has confirmed that AD preferentially targets and impairs primary asymmetric synapses, which are dominated by glutamatergic transmission and constitute the core components of learning and memory-related neural circuits (John and Reddy, 2021). On the specific mechanism level, hyperphosphorylated tau protein accumulates in the postsynaptic density (PSD), impairing the structural integrity of scaffold proteins such as PSD-95. This leads to abnormal membrane localization of glutamate receptors, especially AMPA receptors, thereby significantly reducing neuronal plasticity. Meanwhile, tau pathology indirectly inhibits the transport of brain-derived neurotrophic factor (BDNF) at the presynaptic membrane, resulting in the loss of synaptic trophic support and ultimately forming a vicious cycle of "synaptic dysfunction–neuronal death". Additionally, studies have clearly demonstrated that the extent of loss of such asymmetric synapses is positively correlated with the severity of cognitive decline in AD patients (Reddy and Beal, 2008).

Regarding the site of impairment, synaptic impairment in early AD is predominantly observed in the hippocampus (especially the CA1 region) and the entorhinal cortex (Gao et al., 2025; Igarashi, 2023). Given their critical role in neural circuitry for learning and memory, functional

impairment of these brain regions directly precipitates the early emergence of memory deficits in AD patients. With disease advancement, the impairment spreads progressively to broader brain areas including the prefrontal cortex and temporal cortex, which in turn aggravates cognitive decline (Chen et al., 2022; Yoo et al., 2024). PET imaging analyses of AD patients' brain regions have demonstrated that glucose metabolism in the hippocampus and entorhinal cortex is markedly diminished during the early phases of the disease. This metabolic change indicates that synaptic function has sustained impairment; with disease progression, metabolic perturbations in areas like the prefrontal cortex and temporal cortex worsen progressively, thereby confirming the spreading pattern of synaptic impairment across brain regions (Knopman et al., 2013).

Regarding quantitative and structural alterations, post-mortem analyses have demonstrated that synaptic density in the hippocampus and cortical regions of AD patients is notably lower compared with healthy subjects. Furthermore, the extent of synaptic loss exhibits a marked negative correlation with Mini-Mental State Examination (MMSE) scores, with the key observation that increased synaptic loss aligns with more profound cognitive deficits in patients (Scheff et al., 1990; Terry et al., 1991). Dendritic spines, primary structural substrates of the postsynaptic membrane, likewise undergo a marked decline in density alongside morphological anomalies, including spine head atrophy and spine neck thickening. Such morphological alterations diminish the surface area of the postsynaptic membrane and reduce the enrichment of AMPARs and NMDARs, compromising synaptic signal propagation (Griffiths and Grant, 2023). Meanwhile, the architectural integrity of the presynaptic active zone becomes compromised, while synaptic vesicle counts decrease alongside aberrant localization, thereby ultimately impairing the efficiency of neurotransmitter release (Bose et al., 2024). In AD mouse models, electron microscopy reveals a notable decline in dendritic spine density among hippocampal neurons alongside morphological irregularities. Immunofluorescence staining further demonstrates that the expression of presynaptic SYN and postsynaptic PSD-95 is concurrently diminished. Findings from these animal studies align with the pathological features of human AD patients, thereby offering solid model-based evidence for investigating synaptic damage in AD (Ilina and Linkova, 2023; Zhao et al., 2022).

Synaptic impairment in AD is also tightly linked to specific pathological markers. A β O serves as a critical toxic mediator that driving synaptic impairment, directly interacting with the presynaptic membrane PrPC protein or postsynaptic membrane NMDARs to trigger synaptic dysfunction. Neurofibrillary tangles (NFTs) formed by hyperphosphorylated tau (p-tau) obstruct intracellular material transport in neurons, resulting in insufficient presynaptic neurotransmitter supply. Additionally, they disrupt the dendritic microtubule network and precipitate dendritic spine loss (Peters et al., 2015). Research has demonstrated that intracerebral injection of A β O in mice promptly diminishes synaptic transmission efficiency and impairs LTP, whereas tau silencing partially mitigates A β O-induced synaptic impairment. This observation underscores the synergistic role of A β O and tau in AD-related synaptic damage (Roda et al., 2022; Rolland et al., 2020).

To conclude, the site-specificity (which spreads from the hippocampus and entorhinal cortex to broader brain areas), time-dependent nature (which exacerbates as the disease progresses), and quantifiable structural alterations (characterized by reduced numbers and morphological irregularities) of synaptic impairment in AD are collectively fueled by the synergistic action of A β O and p-tau. These alterations bear direct links to key clinical manifestations of AD, including memory decline and cognitive impairment. These characteristics not only offer a critical gateway for understanding the mechanisms underlying AD progression but also establish a theoretical basis for the subsequent development of AD intervention strategies focused on synaptic protection and carry considerable importance for the basic research and clinical translation of AD.

2.2. Synaptic damage in PD

PD is characterized by resting tremor, muscle rigidity, and bradykinesia as its main symptoms (Li et al., 2025). The pathological hallmark of PD lies in the degenerative loss of midbrain substantia nigra dopaminergic (DA) neurons, with synaptic impairment predominantly localized to dopaminergic synapses and associated neural circuits. These impairments directly underpin the motor and non-motor manifestations of PD (Holmes et al., 2024; Liu et al., 2025; Reddy et al., 2024). Such DA synapses are classified as asymmetric synapses among chemical synapses, which are characterized by a well-developed PSD and primarily distributed on the dendritic spines of striatal neurons. The damage mechanism of these synapses exhibits a multifactorial and synergistic nature: oxidative stress induced by environmental toxins, genetic mutations and other factors directly impairs the function of voltage-gated calcium channels on the presynaptic membrane, resulting in impaired DA neurotransmitter release; following DA depletion, the expression of anchoring proteins between the presynaptic membrane and synaptic vesicles is downregulated, which reduces neurotransmitter transport efficiency, further exacerbates the imbalance in the basal ganglia circuitry, and ultimately induces classic motor symptoms (Mao et al., 2013; Sohrabi et al., 2023).

Loss of dopaminergic synapses is one of the key manifestations of synaptic damage in PD (Radlicka-Borysewska et al., 2024). Synapse counts from midbrain substantia nigra DA neurons innervating the striatum are markedly diminished, and striatal DA levels drop notably. These dual reductions upset the equilibrium of the basal ganglia motor circuit, in turn precipitating PD-related motor manifestations (Gordán-Vélez et al., 2021). Moreover, impairment of noradrenergic synapses in the locus coeruleus and serotonergic synapses within the raphe nucleus is also implicated in PD's non-motor manifestations, such as depression and sleep disorders. This underscores that synaptic impairment in PD is not confined to a single synapse type but encompasses multiple neurotransmitter systems (Buddhala et al., 2015). Research employing dopamine transporter (DAT) imaging of brain regions in PD patients has demonstrated that striatal DAT density is notably decreased. This indicator can act as a surrogate for dopaminergic synapse loss, and the magnitude of this decrease correlates with the severity of motor manifestations—thus furnishing an important basis for the clinical diagnosis and disease progression monitoring of PD (Ba and Martin, 2015).

Presynaptic functional anomalies also exert a pivotal role in synaptic impairment in PD. A key pathological hallmark of PD, α -syn forms Lewy bodies (LBs) and Lewy neurites (LNs) through misfolded aggregation, which compromises the architectural integrity of the presynaptic active zone and impedes the assembly of the SNARE complex. As a critical molecular complex mediating neurotransmitter release, impaired assembly of the SNARE complex results in diminished DA release (Longhena et al., 2018; Mehra et al., 2019). Of greater note, soluble α -synuclein oligomers (α -synO) are also capable of propagating from impaired synapses to healthy ones via a “prion-like propagation” mechanism to broaden the damage scope. This propagative characteristic could constitute a key contributor to the progression of PD (Du et al., 2020). In vitro-cultured dopaminergic neurons show that α -syn overexpression triggers dysregulated assembly of the presynaptic SNARE complex and diminished DA release. Furthermore, introducing soluble α -syn oligomers into the culture medium induces α -syn aggregation and subsequent synaptic impairment in neighboring neurons. Findings from these in vitro experiments offer critical insights into deciphering the mechanisms and propagation patterns of synaptic impairment in PD (Brolin et al., 2023; Ferrari et al., 2023).

Alterations in postsynaptic sensitivity also contribute to the pathological progression of PD. Striatal medium spiny neuron (MSN) expression of DA receptors (D1, D2) is attenuated, and their interaction with PSD scaffold proteins, such as PSD-95, is impaired. Such alterations reduce the efficiency of postsynaptic signal transduction, thereby

compromising the propagation of dopaminergic neural signals (Gerfen et al., 1990; Porras et al., 2012). Research has identified that in PD animal models, striatal MSN mRNA and protein expression of D1 and D2 receptors are concurrently diminished. Co-immunoprecipitation assays further demonstrate that the interaction between these receptors and PSD-95 is attenuated. This reduced interaction impairs the activation of downstream signaling cascades, thereby precipitating PD-related motor dysfunction (Bian et al., 2021; Wei et al., 2017).

To synthesize, synaptic impairment in PD constitutes a multifaceted, multisystemic pathological cascade. Dopaminergic synapse loss serves as the foundational driver of clinical manifestations; α -syn-driven pre-synaptic functional anomalies, including prion-like propagation, act as the pivotal mediator of disease advancement; and dysregulated interactions between postsynaptic DA receptors and scaffold proteins exacerbate synaptic signal transduction deficits. Collectively, these three factors underpin PD's motor and non-motor manifestations. Such features not only elucidate the pathogenic link of “pathological core-synaptic impairment-clinical manifestations” in PD but also offer critical guidance for PD-related basic research and clinical applications.

2.3. Synaptic damage in HD

HD is an autosomal dominant genetic disease caused by CAG repeat expansion in the HTT gene, which leads to the aggregation of mutant huntingtin (mHTT) (Jari et al., 2024). HD presents with choreiform movements, cognitive impairment, and psychiatric manifestations. Synaptic impairment in HD displays distinct regional selectivity in the brain, with predominant involvement of the striatum, and this region-specific damage is tightly linked to HD's clinical symptoms (Waldvogel et al., 2012). Studies have confirmed that the impaired synapses in HD include both excitatory and inhibitory chemical synapses, among which the damage to striatal glutamatergic synapses is the most prominent. The root cause of such damage lies in the interference of mHTT with synaptic cargo transport (Sawant et al., 2021). The core mechanism is axonal transport dysfunction: mHTT binds abnormally to transport-related proteins such as kinesin and HAP1, preferentially inhibiting anterograde transport, which prevents key substances including mitochondria and synaptic vesicles from being delivered to synaptic terminals. The lack of mitochondrial delivery leads to insufficient ATP supply at synapses, while the impaired transport of synaptic vesicles directly causes abnormal neurotransmitter exocytosis. Meanwhile, mHTT enters the synaptic cleft along with vesicles, further inhibiting synaptic plasticity and ultimately resulting in striatal neuronal death (Reddy et al., 2009; Shirendeb et al., 2011).

Early damage to striatal synapses is a key pathological feature of early-stage HD (Raymond, 2017). During the early phase of HD, impairment of striatal glutamatergic synapses, namely cortico-striatal synapses and GABAergic synapses, specifically striato-pallidal synapses, is already evident, with dendritic spine density in striatal neurons showing a notable decline. This impairment compromises the functionality of the basal ganglia motor regulatory circuit. As a critical brain region governing motor control, basal ganglia dysfunction constitutes a primary driver of choreiform movements in HD (Wilton et al., 2023). In mouse HD models, marked depletion of dendritic spines in striatal neurons was detected through Golgi staining. Electrophysiological recordings demonstrated diminished transmission efficiency at cortico-striatal synapses and attenuated inhibitory activity of GABAergic synapses. These animal model-derived observations align with the pathological alterations in human HD patients, laying the groundwork for in-depth investigation of the mechanisms underlying synaptic impairment in HD (André et al., 2011; Spires et al., 2004).

Dysregulated presynaptic neurotransmitter release constitutes another pivotal component of synaptic impairment in HD. mHTT interacts with presynaptic membrane-associated huntingtin-associated protein 1 (HAP1). As a key adapter mediating vesicular transport, this interaction suppresses the interplay between synaptic vesicles and

microtubules, thereby diminishing the synthesis and release of neurotransmitters, including glutamate and GABA (Twelvetrees et al., 2010). Concurrently, mHTT aggregation impairs the function of voltage-gated calcium channels, thereby diminishing Ca^{2+} influx. Ca^{2+} influx serves as a critical initiator of synaptic vesicle fusion; thus, its attenuation further impedes vesicle fusion and aggravates dysregulated neurotransmitter release (Silva et al., 2017). Research has demonstrated that in neurons derived from HD patient-induced pluripotent stem cells (iPSCs), the interaction between mHTT and HAP1 is heightened, vesicular transport is compromised, neurotransmitter release is diminished, and the current density of voltage-gated calcium channels is lowered. These patient-derived cellular observations further validate the role of mHTT in dysregulated presynaptic neurotransmitter release in HD (Mackenzie et al., 2017; Xu et al., 2023). Compromised synaptic plasticity constitutes another pivotal feature of synaptic impairment in HD. In HD patients, striatal LTP induction is compromised, while LTD is dysregulated and exaggerated. This dysregulation undermines the striatum's information processing, learning, and memory capabilities, and is tightly linked to HD-related cognitive impairment (Usdin et al., 1999). The underlying mechanism involves mHTT-mediated diminished CaMKII activity and enhanced AMPAR endocytosis. As a critical kinase mediating LTP induction, diminished CaMKII activity impairs LTP initiation. Conversely, enhanced AMPAR endocytosis reduces the number of postsynaptic membrane receptors, compromising synaptic transmission efficiency (Kung et al., 2007). Striatal brain slices from HD mice showed that high-frequency stimulation was unable to elicit robust LTP, while LTD triggered by low-frequency stimulation was exaggerated. Concurrently, diminished CaMKII activity and enhanced endocytosis of the AMPAR subunit GluA1 were detected. These experimental observations offer direct evidence for deciphering the mechanisms underlying compromised synaptic plasticity in HD (Mandal et al., 2011; Murmu et al., 2015).

Collectively, synaptic impairment in HD constitutes a multi-tiered pathological cascade centered on the striatum and orchestrated by mHTT. Selective impairment of striatal synapses serves as the early underpinnings of motor manifestations; dysregulated presynaptic neurotransmitter release aggravates neural signal transduction deficits; and compromised synaptic plasticity bears direct links to cognitive deficits. These features offer definitive guidance for HD research and intervention, and establish a theoretical framework for the development of early diagnostic approaches and neuroprotective therapies targeting HD.

2.4. Synaptic damage in ALS

ALS is a degenerative disease affecting both upper and lower motor neurons, characterized by progressive muscle weakness and atrophy (Ho et al., 2024; Tian et al., 2024). Synaptic impairment in ALS exhibits distinct site specificity, with predominant involvement of neuromuscular junctions (NMJs) and central motor synapses. Such site-specific damage directly underlies ALS-related motor dysfunction (Vidovic et al., 2024). The NMJ synapses, as specialized asymmetric chemical synapses between motor neurons and skeletal muscle fibers, use acetylcholine as the core neurotransmitter and serve as the final pathway for motor signal transmission (Rodríguez Cruz et al., 2020). Their damage mechanism is centered on excitotoxicity and amplified through multiple intersecting pathways: a surge in glutamate release from astrocytes coupled with postsynaptic calcium overload disrupts the PSD; defects in mitochondrial complex IV trigger an energy crisis, forming a vicious cycle of energy depletion–protein aggregation; mutations in axonal transport genes lead to synaptic nutritional deprivation and subsequent NMJ degeneration (Reddy et al., 2011).

NMJ impairment constitutes one of the pivotal early pathological alterations in ALS. During the early phase of ALS, structural breakdown of NMJs is already evident, alongside diminished acetylcholine (ACh) release at the presynaptic membrane and impaired clustering of

acetylcholine receptors (AChRs) on the postsynaptic membrane (McIntosh et al., 2023). These alterations compromise neuromuscular signal transmission, a vital step mediating muscle contraction, and such compromise precipitates manifestations like muscle weakness and atrophy in ALS patients. ALS animal models and patient muscle biopsies reveal NMJ morphological anomalies, including a diminished number of presynaptic membrane vesicles, flattened postsynaptic membrane folds, and reduced AChR density. Electrophysiological recordings demonstrate lowered amplitude and frequency of endplate potentials. These combined morphological and functional perturbations jointly validate the pivotal role of NMJ impairment in ALS's pathological progression (Ganassi and Zammit, 2022a; Mancuso and Navarro, 2015).

Central motor synaptic impairment also fuels the pathological progression of ALS. Synapse density between cortical motor neurons and spinal anterior horn motor neurons, namely cortico-spinal synapses, diminishes, while presynaptic glutamate release is dysregulated and elevated. Excess glutamate hyperactivates postsynaptic membrane NMDARs, with excitotoxicity acting as a pivotal mediator of neuronal impairment and demise (Feldman et al., 2022). Research has identified that in the spinal cord of ALS mice, cortico-spinal synapse density is diminished, while presynaptic membrane expression of vesicular glutamate transporter 1 (VGLUT1) is elevated. This drives up glutamate concentrations in the synaptic cleft, which further hyperactivates NMDARs and enhances Ca^{2+} influx. Eventually, this impairs spinal anterior horn motor neurons and aggravates motor dysfunction in ALS (Al-Khayri et al., 2024; Feldman et al., 2022).

Furthermore, glial cell-mediated synaptic impairment serves as a key contributor to ALS. Astrocytic downregulation of glutamate transporter 1 (GLT-1) compromises glutamate clearance within the synaptic cleft, driving glutamate accumulation therein and further aggravating excitotoxicity (Feldman et al., 2022). Overactivated microglia release proinflammatory cytokines such as $\text{TNF-}\alpha$ and $\text{IL-1}\beta$, which can directly damage synaptic structures and worsen synaptic damage (Rao et al., 2012). In ALS mouse models, genetic upregulation of astrocytic GLT-1 diminishes glutamate accumulation within the synaptic cleft and mitigates neuronal impairment. Suppression of microglial activation dampens the inflammatory response and preserves synaptic structure and functionality (Guttenplan et al., 2020; Ohashi et al., 2021). These observations offer novel perspectives on ALS treatment, specifically through preserving synaptic integrity by modulating glial cell function.

To integrate, synaptic impairment in ALS arises not from a single mechanism but from the synergistic interplay of three key factors: early structural and functional perturbations of NMJs, excitotoxicity-induced impairment of central motor synapses, and glial cell-mediated indirect impairment. Overall, these factors underpin motor neuron impairment and motor dysfunction. Investigations into these three forms of impairment not only clarify ALS's core pathogenic cascades but also yield critical support for the development of synaptic protection-oriented therapeutic strategies, holding substantial significance for enhancing the prognosis of individuals with ALS.

2.5. Synaptic damage in other neurodegenerative diseases

While Dementia with Lewy Bodies (DLB) and Frontotemporal Dementia (FTD) occur less frequently than the four classic neurodegenerative diseases, their mechanisms of synaptic impairment still bear distinct pathological importance.

DLB exhibits the hallmark of selective synaptic impairment in the hippocampus and cortex. Immunohistochemical analyses demonstrate marked diminution of synaptic density in the hippocampus of DLB patients, with reduced expression of the postsynaptic density protein PSD-95 correlating positively with the pace of cognitive score deterioration (Bereczki et al., 2018). The core mechanism lies in the imbalanced interaction between α -syn and vesicle-associated membrane protein 2 (VAMP2) on the presynaptic membrane (Sun et al., 2019). Under physiological contexts, VAMP2 associates with α -syn through

charge-based residues, yielding co-aggregates. This interaction fosters the assembly of the soluble N-ethylmaleimide-sensitive factor attachment protein receptor complex (SNARE) and suppresses α -syn oligomer formation (Wang et al., 2024a). Under pathological contexts, VAMP2 depletion triggers dysregulated α -syn aggregation. This not only impedes synaptic vesicle fusion but also heightens neuroinflammation through activation of the p38 α inflammation pathway, worsening synaptic degeneration (Germann and Alam, 2020a; Yoo et al., 2022). Unlike the localized impairment of dopaminergic synapses in PD, α -syn toxicity in DLB exerts broader effects on cortical networks.

Synaptic damage in FTD exhibits significant genotype specificity, with typical manifestations of synaptic loss in the frontal and temporal cortices (Dando et al., 2024; Gelon et al., 2022). Dysregulated aggregation of TAR DNA-binding protein 43 (TDP-43) serves as the core pathogenic driver, disrupting post-transcriptional regulation through binding to the 3' untranslated region (3'UTR) of synapse-related mRNAs (Ayala et al., 2011). Research has identified that in patients with frontotemporal lobar degeneration with TDP-43 inclusions (FTLD-TDP), TDP-43 aggregation triggers dysregulated accumulation of mRNA encoding the synaptic protein neuronal pentraxin 2 (NPTX2), with this pathway being unique to humans (Hruska-Plochan et al., 2024). Imaging analyses demonstrate that patients with semantic variant FTD arising from granulin (GRN) gene mutations exhibit marked diminution of synaptic density in the left frontal, temporal, and parietal lobes, with involvement extending even to the contralateral cerebellar hemisphere. Of note, the scope of synaptic loss outpaces structural brain atrophy (Rodriguez-Vieitez et al., 2025; Salzano et al., 2025). Fluid biomarker analyses uncover increased concentrations of neurogranin (Ng) in the CSF of FTD patients with microtubule-associated protein tau (MAPT) gene mutations, a pattern not observed in patients harboring C9orf72 or GRN mutations. This indicates that synaptic impairment pathways vary by genetic subtype (Sogorb-Esteve et al., 2022). These findings provide a molecular basis for the early diagnosis and subtype-specific intervention of FTD.

While distinct neurodegenerative diseases possess unique pathological hallmarks, they all coincide with diminished synaptic quantity, structural impairment, and functional perturbations throughout their pathological progression. Furthermore, the localization, cell type and severity of synaptic damage correlate directly with the clinical manifestations of these conditions, and this shared characteristic offers a valuable avenue for investigating the common pathological mechanisms underlying neurodegenerative diseases (Fig. 1, Table 1).

3. Molecular mechanisms of synaptic damage in neurodegenerative diseases

This section will systematically review five core molecular mechanisms driving synaptic dysfunction, including pathological protein toxicity, synaptic protein homeostasis dysregulation, excitotoxicity/calcium imbalance, oxidative stress/inflammation, and neurotrophic factor deficiency, with detailed elaboration on how disease-specific pathological proteins ($A\beta$ oligomers, α -synuclein aggregates, mutant huntingtin, etc.) orchestrate these cascades in neurodegenerative diseases.

3.1. Synaptic toxic effects of pathological proteins

In the pathological progression of neurodegenerative diseases, different diseases tend to have disease-specific pathological proteins, and abnormal alterations of these proteins are among the core biological hallmarks of the diseases. Once these pathological proteins undergo abnormal aggregation at synapses, they induce synaptic damage through direct or indirect effects. They serve as the initiating factors of synaptic damage and play a crucial role in driving the occurrence and progression of the diseases (Fig. 2).

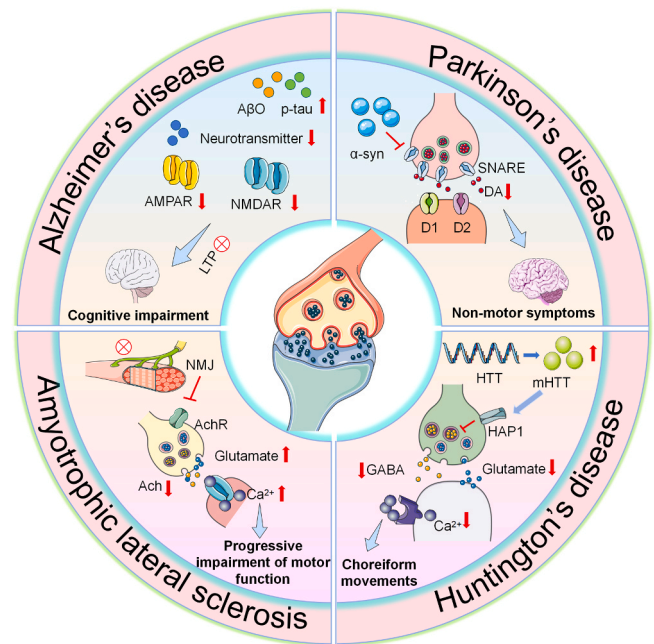


Fig. 1. Pathological characteristics of synaptic damage in neurodegenerative diseases. In AD, $A\beta$ O and p-tau accumulate, accompanied by reduced neurotransmitter levels, impaired AMPAR/NMDAR function, and suppressed LTP, ultimately leading to cognitive impairment. In PD, centered on α -syn aggregation and abnormal SNARE complex function, there is a concurrent reduction in DA secretion and dysregulation of D1/D2 receptor signaling, which corresponds to the onset of non-motor symptoms. In HD, mHTT impairs HAP1 function, leading to decreased GABA levels, glutamate imbalance, and Ca^{2+} homeostasis disruption, manifesting as choreiform movements. In ALS, the NMJ is affected, with abnormal AChR, accompanied by dysregulation of ACh and glutamate signaling as well as Ca^{2+} overload, ultimately resulting in progressive impairment of motor function. AD (alzheimer's disease), $A\beta$ O (β -amyloid oligomers), p-tau (phosphorylated tau), AMPAR (alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor), NMDAR (N-Methyl-D-aspartate receptor), LTP (long-term potentiation), PD (parkinson's disease), α -syn (α -synuclein), SNARE (soluble NSF attachment protein receptor), DA (dopamine), HD (huntington's disease), mHTT (mutant huntingtin protein), HAP1 (huntingtin associated protein 1), GABA (γ -aminobutyric acid), ALS (amyotrophic lateral sclerosis), NMJ (neuromuscular junction), AChR (acetylcholine receptors), accompanied by dysregulation of ACh (acetylcholine). Figure created with PowerPoint.

3.1.1. Synaptic toxicity mechanisms of $A\beta$

$A\beta$ O are the primary form through which $A\beta$ exerts synaptic toxicity, and they induce synaptic damage via multiple pathways (Forloni, 2023). $A\beta$ O engages with key molecules on the presynaptic membrane, perturbing the equilibrium of neurotransmitter release by dysregulating the activation of calcium channels. $A\beta$ O selectively interacts with the prion protein (PrPC) on the presynaptic membrane, and this interaction triggers the activation of Src family kinases (e.g., Fyn). The activated Fyn then induces hyperactivation of voltage-gated calcium channels (CaV2.1), resulting in aberrant elevation of Ca^{2+} influx across the presynaptic membrane (Balducci et al., 2025). Excessive Ca^{2+} influx triggers two problems (De Mario et al., 2015; Wegierski and Kuznicki, 2018): For one thing, it induces excessive neurotransmitter exocytosis from synaptic vesicles; for another, it may trigger premature exhaustion of the synaptic vesicle reserve. Over time, this perturbs the dynamic equilibrium of neurotransmitter release. This pathological cascade has been well-documented in primary hippocampal neuron cultures. Following $A\beta$ O administration, the abundance of PrPC-Fyn complexes rises, CaV2.1 phosphorylation is upregulated, and presynaptic Ca^{2+} influx is markedly potentiated. Ultimately, the synaptic vesicle reserve becomes exhausted, and neurotransmitter release undergoes a transient surge

Table 1
Affected cell types and core brain regions in major neurodegenerative diseases.

Disease name	Affected cell types	Core brain sites	Ref.
AD	Pyramidal cells in the hippocampal CA1 region, entorhinal cortex neurons, cortical pyramidal cells.	Hippocampus, entorhinal cortex, prefrontal cortex	(Chen et al., 2022; Gao et al., 2025; Igarashi, 2023)
PD	Dopaminergic neurons in the substantia nigra pars compacta, medium spiny neurons in the striatum.	Substantia nigra, striatum, locus coeruleus	(Ba and Martin, 2015; Buddhala et al., 2015; Gordián-Vélez et al., 2021)
HD	Medium spiny neurons in the striatum, cortical neurons.	Striatum, cerebral cortex.	(André et al., 2011; Raymond, 2017; Wilton et al., 2023)
ALS	Spinal motor neurons, cortical motor neurons, Schwann cells associated with the NMJ.	Spinal cord, motor cortex, neuromuscular junction	(Ganassi and Zammit, 2022b; McIntosh et al., 2023; Vidovic et al., 2024)
DLB	Hippocampal neurons, cortical neurons, dopaminergic neurons.	Hippocampus, cerebral cortex, substantia nigra	(Bereczki et al., 2018; Germann and Alam, 2020b)
FTD	Prefrontal/temporal cortical neurons, presynaptic neurons	Prefrontal cortex, temporal cortex	(Geloni et al., 2022)

followed by sustained decline—this directly validates AβO-mediated presynaptic dysfunction (Larson et al., 2012; Peters et al., 2015). Beyond its actions on the presynaptic membrane, AβO also engages directly with

postsynaptic membrane NMDARs, triggering pro-apoptotic pathways through augmented Ca²⁺ influx and impairing postsynaptic neurons (Prihodko et al., 2024). Notably, AβO displays selective binding to NMDARs, showing a distinct preference for GluN2B-subunit-containing NMDARs. This interaction markedly amplifies receptor-mediated Ca²⁺ influx, and elevated Ca²⁺ then triggers activation of downstream pro-apoptotic pathways including caspase-3 and JNK, inducing toxic impairment of postsynaptic neurons (Rönicke et al., 2011). This mechanism has been corroborated in the AD APP/PS1 transgenic mouse model. In the hippocampus of these mice, AβO-GluN2B colocalization is markedly elevated. Upon inhibition of GluN2B activity, Ca²⁺ influx was substantially diminished, and the apoptosis rate of postsynaptic neurons concomitantly declined. This finding indicates that targeted suppression of GluN2B activity may serve as a promising therapeutic strategy to mitigate AβO-mediated synaptic impairment (Pousinha et al., 2017).

AβO-mediated postsynaptic membrane impairment also presents as disrupting the interplay between PSD-95 and NMDARs, resulting in receptor destabilization and endocytic degradation (Liu et al., 2010). PSD-95 serves as a pivotal scaffold protein in the postsynaptic density, with its central role being the binding of NMDARs to sustain stable receptor expression and proper membrane localization on the postsynaptic membrane (Lin et al., 2004). However, AβO directly disrupts this interplay, diminishing the binding capacity between the two. Co-immunoprecipitation assays demonstrate that in AβO-treated hippocampal neurons, the binding levels of PSD-95 to both GluN2A and GluN2B (two NMDAR subunits) were notably reduced. Receptor dissociation from scaffold proteins induces NMDAR destabilization on the postsynaptic membrane, with subsequent endocytic degradation by the cell that diminishes surface NMDAR expression. Reduced receptor abundance directly compromises postsynaptic signal transduction

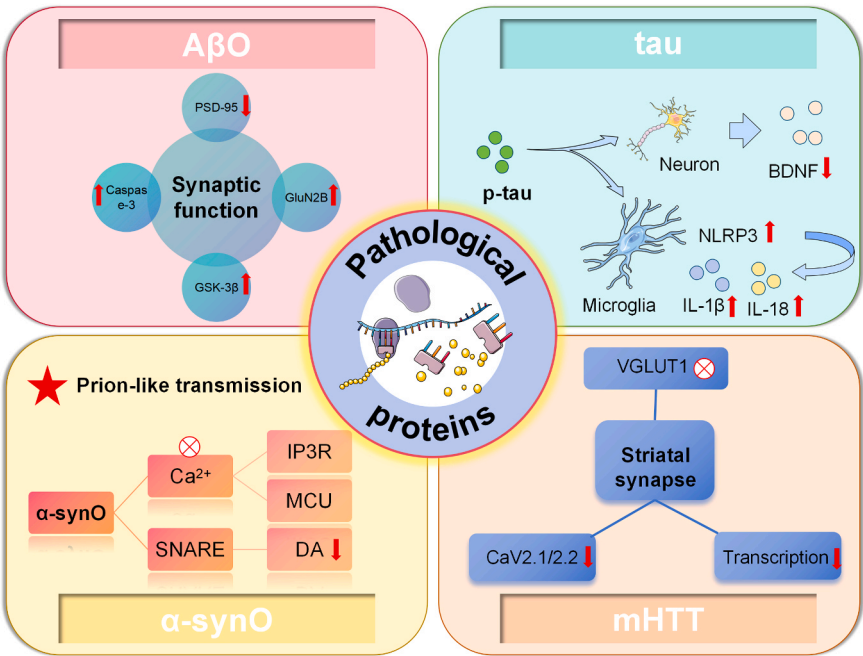


Fig. 2. The pathogenic mechanisms of core pathological proteins in neurodegenerative diseases. AβO: It downregulates synaptic molecules such as PSD-95 and GluA2, activates Caspase-3 (apoptotic pathway) and GSK-3β (kinase pathway), and directly impairs synaptic function. It is one of the core drivers of synaptic pathology in AD. Pathogenic cascade of tau protein: p-tau induces neuronal damage, accompanied by downregulation of BDNF; it also activates the NLRP3 inflammasome in microglia, promotes the release of inflammatory cytokines such as IL-1β, and amplifies the neuroinflammatory response. Role of α-synO: It has prion-like transmission properties, can interfere with the function of the SNARE complex, disrupt calcium homeostasis (abnormalities in IP3R and MCU), and ultimately reduce DA secretion, participating in the neurotransmitter dysregulation process in PD. Impact of mHTT: It inhibits VGLUT1 function and impairs striatal synapses, while interfering with Cav2.1/2.2 channels and transcription processes. It is a key mechanism underlying neuronal dysfunction in HD. AβO (β-amyloid oligomer), PSD-95 (postsynaptic density protein 95), GSK-3β (glycogen synthase kinase 3 beta), p-tau (phosphorylated tau), AD (alzheimer's disease), BDNF (brain-derived neurotrophic factor), NLRP3 (NOD-like receptor family pyrin domain containing 3), IL-1β (interleukin-1 beta), α-synO (α-synuclein oligomer), SNARE (soluble NSF attachment protein receptor), IP3R (inositol 1,4,5-trisphosphate receptor), MCU (mitochondrial calcium uniporter), DA (dopamine), mHTT (mutant huntingtin protein), PD (parkinson's disease), VGLUT1 (vesicular glutamate transporter 1), HD (huntington's disease). Figure created with PowerPoint.

efficiency, further impairing synaptic transmission capacity (Roselli et al., 2005).

A β O-mediated synaptic impairment also encompasses intracellular signaling pathways, driving presynaptic tau phosphorylation via activation of glycogen synthase kinase 3 β (GSK-3 β) and perturbing presynaptic material transport (Takashima, 2006). The precise mechanism operates as follows: A β O suppresses the PI3K/Akt signaling pathway, alleviating Akt's phosphorylation-dependent suppression of GSK-3 β . Under physiological circumstances, Akt inactivates GSK-3 β via phosphorylation; pathway suppression elevates GSK-3 β activity markedly. Activated GSK-3 β selectively phosphorylates presynaptic tau protein, with preferential targeting of key residues including Ser396 and Thr231 (Fan et al., 2022). Following phosphorylation, tau's tubulin-binding capacity is substantially diminished, triggering microtubule depolymerization. Microtubules act as "tracks" for presynaptic cargo transport, and their depolymerization directly perturbs the presynaptic membrane's cargo transport function, compromising the proper trafficking of key components, including synaptic vesicles and receptors, ultimately endangering the structural integrity and functional preservation of the presynaptic membrane (Fan et al., 2022; Wu et al., 2021).

3.1.2. Synaptic toxicity mechanisms of tau

Under physiological circumstances, tau protein sustains microtubule stability via tubulin binding—a process pivotal to neuronal cargo transport and structural integrity (Chinnathambi et al., 2025; Duggal et al., 2025). However, p-tau forming oligomers or NFTs abrogate their physiological function and induce synaptic toxicity primarily through three mechanisms (Singh et al., 2025).

For one thing, p-tau perturbs the microtubule architecture of neurons and impedes intracellular cargo transport, severing the source supply of key components to synapses. The central role of normal tau lies in binding to microtubules and sustaining their stability, yet p-tau fails to retain this binding capacity, triggering depolymerization of the microtubule network within neuronal axons and dendrites (Li et al., 2007). Microtubules function as the primary transport channels for intracellular cargo in neurons, tasked with trafficking key components, including neurotransmitters and neurotrophic factors (e.g., BDNF), from neuronal cell bodies to synaptic terminals. Their depolymerization directly obstructs this trafficking pathway (Biswas and Das Sarma, 2014). Mouse hippocampal neuron experiments show that mutant tau (P301L) overexpression elicits marked diminution of microtubule density and substantial decline in the trafficking rate of BDNF from neuronal cell bodies to synaptic terminals (Kandimalla et al., 2016). This trafficking impairment triggers inadequate provision of neurotrophic factors to the presynaptic membrane, directly compromising synaptic survival integrity and functional preservation.

Beyond perturbing cargo transport, p-tau-derived oligomers can also diffuse directly to the postsynaptic membrane and compromise the structural integrity of the PSD. These oligomers can traverse the synaptic cleft to reach the postsynaptic membrane and selectively interact with core scaffold proteins within the PSD (e.g., PSD-95, Shank3) (Rawat et al., 2022). These PSD scaffold proteins serve as core molecules upholding the spatial architecture of the PSD and anchoring neurotransmitter receptors; their interaction with p-tau directly perturbs the PSD's inherent structure (Shen et al., 2023). High-resolution images derived from cryo-electron microscopy analyses reveal that p-tau oligomers may intercalate into the PSD's layered architecture, eliciting elevated PSD thickness and diminished density (Sun et al., 2023). This structural alteration further undermines postsynaptic membrane AMPAR aggregation. As core receptors underpinning excitatory synaptic transmission, diminished AMPAR aggregation directly diminishes synaptic transmission efficiency and compromises synaptic signal transduction capacity (Jung et al., 2023).

Tau-mediated synaptic impairment also elicits inflammatory responses through microglial engagement, establishing an "indirect damage pathway". Microglia can identify and phagocytose p-tau via

surface scavenger receptors (e.g., CD36). Following phagocytosis, p-tau engages the NLRP3 inflammasome within microglia, a core signaling complex that governs the release of proinflammatory cytokines (Liang et al., 2022). NLRP3 activation fuels the robust secretion of proinflammatory cytokines, including IL-1 β and IL-18, and these cytokines then spread to impair surrounding synapses (Ising et al., 2019). In brain sections from AD patients, microglia surrounding NFTs (primarily composed of p-tau) are generally in an activated state, with significantly increased expression of NLRP3 and IL-1 β (Ahmed et al., 2017). Subsequent experiments confirmed that inhibiting NLRP3 activity can significantly reduce microglia-mediated synaptic damage (Jung et al., 2025). This finding not only validates that p-tau exerts indirect synaptic damage via inflammatory cascades but also lays a valuable theoretical foundation for anti-inflammatory therapies in AD.

3.1.3. Synaptic toxicity mechanisms of α -syn

α -syn is a soluble protein residing at the presynaptic membrane. Under physiological circumstances, it contributes to synaptic vesicle recycling and refilling, serving as a key mediator of synaptic vesicle cycling and neurotransmitter release (Gcwensa et al., 2024; Wang et al., 2024b). However, when α -syn undergoes aberrant aggregation into α -synO or LBs, it elicits substantial synaptic toxicity, emerging as a central mediator of synaptic impairment in PD. Its toxic modalities encompass four primary facets.

For one thing, α -synO targets and disrupts the normal cycling of synaptic vesicles, encompassing two core steps: "fusion impairment" and "endocytosis blockage". During synaptic vesicle fusion with the presynaptic membrane, α -synO selectively interacts with core components of the SNARE complex (e.g., Syntaxin-1A, VAMP2, SNAP-25). This complex serves as the core molecular machinery underpinning synaptic vesicle-membrane fusion, and its interaction directly compromises the complex's assembly efficiency (Lou et al., 2017). In an *in vitro* reconstitution system, α -synO binds to Syntaxin-1A with a dissociation constant (K_d) of 120 nM, markedly lower than the 800 nM observed for native α -syn. These findings confirm that α -synO exhibits higher binding affinity for Syntaxin-1A, thereby diminishing the assembly efficacy of the SNARE complex by 40–60 percent and directly compromising neurotransmitter release (Choi et al., 2013). During synaptic vesicle endocytic recycling, α -synO also interacts with the clathrin adaptor protein AP-2. AP-2 serves as a core molecule underpinning clathrin-mediated endocytosis (CME) of synaptic vesicles, with this interaction obstructing the endocytic process (McCormack et al., 2019). In substantia nigra neurons of A53T α -syn transgenic PD mice, AP-2 and α -syn exhibit markedly elevated colocalization, while synaptic vesicle endocytic rate is diminished. These alterations ultimately elicit depletion of the synaptic vesicle pool and a 35 percent diminution in DA release, further aggravating dopaminergic nerve dysfunction in PD patients (Koprich et al., 2010).

Beyond perturbing synaptic vesicle cycling, α -synO also deranges intracellular calcium homeostasis and elicits synaptic protein degradation via aberrant regulation of calcium flux. Its modulation of calcium equilibrium operates primarily via two routes: For one aspect, α -synO selectively interacts with inositol 1,4,5-trisphosphate receptors (IP3Rs) on the endoplasmic reticulum (ER), inducing receptor activation and driving robust efflux of ER-sequestered Ca²⁺ into the cytoplasmic compartment (Yamamoto et al., 2019). On the other hand, it suppresses the activity of mitochondrial calcium uniporters (MCU), diminishing mitochondrial Ca²⁺ uptake from the cytoplasmic compartment. Collectively, these dual effects elicit excessive elevation of free Ca²⁺ concentration in the synaptic region (Ramezani et al., 2023). Research findings confirm that in α -synO-exposed dopaminergic neurons, ER Ca²⁺ release doubles, while mitochondrial Ca²⁺ uptake is reduced by half (Ramezani et al., 2023). Excess Ca²⁺ elicits activation of calpain, a calcium-dependent protease that selectively degrades core synaptic proteins, including PSD-95 and microtubule-associated protein 2 (Kovacs et al., 2021). This directly damages the structural integrity and

functional stability of synapses.

More critically, α -synO exhibits prion-like transcellular propagation properties, which allow the scope of synaptic damage to expand continuously and drive disease progression (Kuo et al., 2025). α -synO may be secreted into the extracellular space via impaired synaptic clefts and internalized by adjacent healthy neurons through endocytosis. Upon entry into recipient neurons, α -synO functions as a “seed” to trigger conformational changes in normally folded α -syn within cells, driving its misfolding and aggregation into novel α -synO. This in turn compromises the synapses of these neurons, establishing a “synaptic propagation chain” (Desplats et al., 2009). This propagation mechanism not only accounts for the spread of synaptic damage in PD from localized regions to widespread brain areas but also lays a valuable theoretical underpinning for deciphering disease progression.

3.1.4. Synaptic toxicity mechanisms of mHTT

HD arises from CAG repeat expansion in the huntingtin (HTT) gene, triggering the aggregation of mHTT. The synaptic toxicity of mHTT displays distinct brain region selectivity, predominantly impairing the striatum, with this selectivity linked to the high susceptibility of striatal neurons to mHTT (Mätlik et al., 2024). Its toxic mechanisms mainly include three aspects.

For one thing, mHTT selectively interacts with and disrupts intracellular vesicular transport pathways in neurons, abrogating the delivery of essential molecules to the presynaptic membrane. It selectively engages with HAP1 – a central adaptor molecule in vesicular trafficking that primarily mediates vesicle-microtubule interactions, facilitating vesicular transport along microtubules to the presynaptic membrane. Following this interaction, HAP1's capacity to mediate vesicle-microtubule crosstalk becomes compromised, thereby impeding the vesicular transport process outright (Mackenzie et al., 2017). This pathological impact is particularly salient in striatal neurons differentiated from iPSCs derived from HD patients: binding frequency between mHTT and HAP1 triples relative to baseline levels, while the trafficking velocity of VGLUT1 – a glutamate-trafficking transporter – declines by six tenths. This trafficking impairment results in insufficient glutamate trafficking to the presynaptic membrane and reduced glutamate exocytosis, thus undermining the signal propagation efficacy of excitatory synapses (Chen et al., 2023).

Beyond perturbing vesicular transport, mHTT also suppresses the function of voltage-gated calcium channels localized to the presynaptic membrane, diminishing Ca^{2+} influx, thereby impeding neurotransmitter release (Miranda et al., 2019). The distinctive polyglutamine (polyQ) domain of mHTT engages directly with the β subunit of voltage-gated calcium channels; this direct interaction induces conformational changes in the channels and markedly diminishes the open probability of $\text{CaV}2.1/2.2$ channels (Meir et al., 2000). As the primary channels mediating Ca^{2+} influx at the presynaptic membrane, reduced open probability of $\text{CaV}2.1/2.2$ channels directly precipitates diminished Ca^{2+} influx. Within striatal brain slices of HD model mice (R6/2), the current density of $\text{CaV}2.1$ channels is markedly attenuated. Inadequate Ca^{2+} influx impairs synaptic vesicle-membrane fusion efficiency, thereby further suppressing neurotransmitter release. This constitutes a “double hit” of “compromised material trafficking+insufficient calcium influx”, aggravating presynaptic dysfunction (Klapstein et al., 2001).

mHTT-induced synaptic damage also encompasses postsynaptic signal transduction, dampening the expression of key signaling molecules by disrupting transcriptional control (Sari, 2011). mHTT is capable of translocating to the neuronal nucleus and engaging with transcription factors such as CREB and SP1, suppressing their functional activity to alter the transcription of postsynaptic signal-related genes (Chaturvedi et al., 2012). Notably among these genes, dopamine- and cAMP-regulated phosphoprotein of 32 kDa (DARPP-32) stands out as one of the most prominently impacted molecules. DARPP-32 serves as a core component of dopamine receptor-mediated signaling pathways and plays a pivotal role in dopamine signal processing within postsynaptic

neurons. In the striatum of R6/2 mice, DARPP-32 expression is markedly diminished. Its downregulation attenuates PKA signaling pathway activation triggered by dopamine receptors, diminishes postsynaptic neuronal sensitivity to dopamine, and impedes the normal transmission and processing of dopaminergic signals, thereby disrupting postsynaptic function (Choi et al., 2024).

3.2. Dysregulation of synaptic protein homeostasis

3.2.1. Decreased synthesis of synaptic proteins

Neurons harbor a “local synaptic protein synthesis” mechanism whereby synaptic mRNA can rapidly generate synaptic proteins such as PSD-95 and AMPAR through local translation. This mechanism facilitates rapid adaptation to synaptic plasticity demands and plays a pivotal role in preserving the flexibility of synaptic function (Choi et al., 2024). However, in neurodegenerative diseases, pathological proteins impede mRNA transport or translation initiation via diverse pathways, resulting in impaired synthesis of synaptic proteins.

In HD, mHTT mainly compromises excitatory synaptic transmission by suppressing mRNA translation of key presynaptic proteins. mHTT can bind to the translation initiation factor eIF4E – a key mediator of translation initiation that binds to the 5'mRNA cap structure. This interaction blocks eIF4E from engaging with the 5'mRNA cap structure, thereby suppressing translation initiation (Creus-Muncunill et al., 2019). In synapses of the striatum in HD mice, the binding abundance of eIF4E to VGLUT1 mRNA is markedly reduced, and VGLUT1 protein expression is substantially diminished. This leads to reduced glutamate release, compromising excitatory synaptic transmission (Shigeri and Shimamoto, 2003). Beyond HD, ALS encompasses various RNA metabolism disorders driven by pathogenic protein aberrations. Among these, aberrant aggregation of TDP-43 directly obstructs mRNA transport of key postsynaptic proteins. In ALS, aggregated TDP-43 forms cytoplasmic inclusions that bind to AChR mRNA. This binding blocks AChR mRNA translocation from the cell body to the NMJ (Armstrong and Drapeau, 2013). In muscle biopsy samples from ALS patients, AChR mRNA levels at the NMJ are notably reduced, and AChR protein expression is markedly decreased. This disrupts NMJ signal transmission and impairs neuromuscular function (Palma et al., 2011). Another pathogenic mechanism linked to RNA metabolism in ALS is the suppression of synaptic protein transcription by dysregulated nuclear localization of fused in sarcoma (FUS) protein. FUS is an RNA-binding protein that is constitutively localized in the nucleus and contributes to the regulation of gene transcription (Sahadevan et al., 2021). In ALS, FUS is exported from the nucleus and accumulates in the cytoplasm, precluding its nuclear re-entry to regulate synaptic protein transcription (Vance et al., 2013). In the hippocampus of FUS knockout mice, PSD-95 mRNA transcription levels are markedly reduced, and PSD-95 expression at the postsynaptic membrane is diminished, leading to decreased synaptic transmission efficiency (Wang et al., 2023a). This provides additional confirmation of FUS's critical role in synaptic protein transcription.

3.2.2. Impairment of synaptic protein degradation

The ubiquitin-proteasome system (UPS) and autophagy-lysosome system constitute the primary degradation pathways for synaptic proteins. The proper functionality of these dual systems directly dictates whether synaptic proteins undergo timely degradation to preclude abnormal accumulation (Tai and Schuman, 2008). In neurodegenerative diseases, these two systems exhibit functional aberrations, resulting in synaptic protein accumulation and consequent synaptic impairment.

In AD, $\text{A}\beta\text{O}$ serve as key mediators disrupting protein degradation balance, primarily inducing synaptic impairment by suppressing proteasomal activity. $\text{A}\beta\text{O}$ can bind to the 20S core particle of the proteasome, the central component that underpins the proteasome's degradative activity. This interaction abrogates the proteasome's peptidase function (Davidson et al., 2025). In hippocampal synapses of

APP/PS1 mice, the chymotrypsin-like activity of the proteasome is markedly reduced, slowing the degradation rate of SYN and post-synaptic PSD-95. This leads to substantial accumulation of these proteins, which interfere with normal synaptic function and elicit synaptic impairment (Torres et al., 2012). In contrast, α -syn in PD primarily perturbs protein homeostasis by inhibiting autophagosome formation, thereby triggering synaptic toxicity. α -syn can interact with the autophagy-related protein LC3, a key molecule in autophagosome biogenesis, with the conversion of LC3-I to LC3-II serving as a key hallmark of autophagosomal membrane formation. The binding between α -syn and LC3 obstructs the conversion of LC3-I to LC3-II, suppressing autophagosomal membrane assembly (Xu et al., 2025). In the substantia nigra of A53T α -syn transgenic mice, the LC3-II/LC3-I ratio is significantly diminished, the number of autophagosomes is notably reduced, and synaptic α -syn accumulation is markedly elevated—further exacerbating synaptic toxicity (Pupyshev et al., 2018). HD exhibits a more distinctive pattern of protein degradation dysfunction: mHTT primarily induces synaptic impairment by compromising lysosomal function, forming a “damage-accumulation” vicious cycle. mHTT can bind to the lysosomal membrane protein LAMP1, a key molecule that maintains lysosomal membrane stability. This binding enhances lysosomal membrane permeability, causing leakage of lysosomal enzymes such as cathepsin B (Sun et al., 2024a). In striatal synapses of R6/2 mice, lysosomal enzyme leakage is doubled, leading to aberrant degradation of synaptic proteins such as Shank3, while undegraded mHTT accumulates within lysosomes. This establishes a vicious cycle that continuously aggravates synaptic impairment (Hering et al., 2015).

3.3. Excitotoxicity and calcium homeostasis dysregulation

3.3.1. Abnormal metabolism of excitatory neurotransmitters

Glutamate levels in the synaptic cleft are coordinately modulated by two processes: release and clearance. Under physiological conditions, these two processes exist in equilibrium, sustaining glutamate concentration within an optimal range. In neurodegenerative diseases, however, this equilibrium is perturbed, resulting in glutamate accumulation that lays the groundwork for excitotoxicity.

From the perspective of presynaptic glutamate release, distinct neurodegenerative diseases elicit abnormally elevated glutamate release via unique mechanisms. In AD, A β O activates presynaptic mGluR5 receptors and induces glutamate exocytosis from synaptic vesicles through the PLC-IP₃ pathway (He et al., 2019). In PD, by contrast, α -synO suppresses DA release: DA normally restrains striatal glutamatergic neurons, so reduced DA secretion alleviates this inhibitory effect and enhances glutamate release. Experimental evidence directly confirms aberrant presynaptic glutamate release in these diseases: synaptic cleft glutamate concentration is notably increased in hippocampal brain slices from APP/PS1 mice (Minkeviciene et al., 2008); while glutamate release is markedly elevated in the striatum of A53T mice (Unger et al., 2006). These findings further validate the phenomenon of excessive presynaptic glutamate release in neurodegenerative diseases.

Beyond release abnormalities, impairment of synaptic cleft glutamate clearance mechanisms exacerbates glutamate accumulation, creating a “double dilemma” of “excessive release+insufficient clearance”. Under normal physiological conditions, astrocytes take up the majority of synaptic cleft glutamate via the surface glutamate transporter GLT-1, also known as EAAT2. GLT-1’s transporter function serves as a key safeguard for maintaining synaptic cleft glutamate concentration stability (Zhou and Sutherland, 2004). In neurodegenerative diseases, however, GLT-1 expression or function is impaired. In ALS, mutant SOD1 binds to GLT-1 and suppresses its transport activity (Trotti et al., 1999); in AD, A β O triggers microglial activation to secrete TNF- α , which diminishes GLT-1 expression in astrocytes (Mookherjee et al., 2011). In the spinal cord of SOD1G93A mice, an ALS model, GLT-1 transport efficiency is significantly reduced (Pardo et al., 2006); in the cortex of APP/PS1 mice, GLT-1 expression is notably diminished,

leading to a marked prolongation of glutamate clearance half-life in the synaptic cleft and further exacerbating glutamate buildup (Gao et al., 2020).

3.3.2. Overactivation of NMDAR and Ca²⁺ influx

Following glutamate buildup in the synaptic cleft, it hyperactivates NMDARs at the postsynaptic membrane, notably those harboring the GluN2B subunit. These receptors exhibit high affinity for glutamate and facilitate enhanced Ca²⁺ influx upon activation, establishing them as the primary source of Ca²⁺ entry during excitotoxicity (Yu et al., 2023). While the mechanisms underlying NMDAR hyperactivation vary across neurodegenerative diseases, they all converge on massive Ca²⁺ influx as a common endpoint.

In AD, p-tau serves as the key mediator of aberrant NMDAR activation, specifically inducing synaptic impairment by elevating the receptor’s open probability (Park et al., 2020). p-tau is capable of interacting with the GluN2A subunit of NMDARs, and this interaction extends the receptor’s open duration. In hippocampal neurons from AD patients, colocalization of p-tau and GluN2A is increased; NMDAR open probability is markedly elevated, and Ca²⁺ influx is substantially enhanced. This excessive Ca²⁺ entry damages postsynaptic neurons (Yeung et al., 2021). In contrast to the direct modulation by p-tau in AD, aberrant NMDAR activity in PD primarily manifests as enhanced receptor sensitivity in striatal MSNs. In the striatum of PD mice, the binding abundance of NMDARs to PSD-95 in MSNs is elevated. PSD-95 functions as a scaffold protein that sustains the clustering and stability of NMDARs at the postsynaptic membrane, and augmented binding between the two reinforces receptor aggregation, boosting the receptors’ glutamate sensitivity. Even modest glutamate levels can activate the receptors and mediate Ca²⁺ influx, exacerbating Ca²⁺ overload (Porras et al., 2012). ALS is distinguished by NMDAR hyperactivation in spinal cord anterior horn motor neurons, a phenomenon closely linked to alterations in receptor subunit expression and phosphorylation status. In the spinal cord of ALS mice, motor neuron expression of the NMDAR GluN2B subunit is markedly upregulated, alongside elevated receptor phosphorylation. Both GluN2B upregulation and increased receptor phosphorylation enhance NMDAR activity, leading to augmented Ca²⁺ influx. This engages downstream toxic signaling cascades and impairs motor neuron synapses (Mallozzi et al., 2018).

3.3.3. Ca²⁺ overload-mediated synaptic damage effects

Following massive Ca²⁺ entry, synaptic impairment is exacerbated via multiple interconnected pathways, primarily encompassing three facets: activation of proteases, promotion of reactive oxygen species (ROS) generation, and disruption of mitochondrial function. These overlapping and synergistic effects converge to disrupt synaptic architecture and abrogate functional integrity.

For starters, Ca²⁺ overload directly activates calcium-dependent proteases. Among these enzymes, calpain and Caspase-3 serve as key effectors mediating synaptic damage, acting through distinct dimensions: calpain via “structural destruction” and Caspase-3 via “apoptosis induction” (D’Amelio et al., 2011; Liu et al., 2008). Calpain degrades synaptic structural proteins such as PSD-95, a key regulator of postsynaptic density architecture, and tubulin, a scaffold for neuronal cytoskeletal integrity, directly compromising the physical structure of synapses (Ayuso-Dolado et al., 2021). Caspase-3 primarily triggers postsynaptic neuronal apoptosis, depriving synapses of their essential neuronal support (Cowan et al., 2001). This mechanism is particularly pronounced in hippocampal tissue from AD mice: calpain activity is markedly elevated, leading to substantial PSD-95 degradation (Saito et al., 2016); Caspase-3 activity is further augmented, resulting in a notable increase in postsynaptic neuronal apoptosis rate (Niu et al., 2010). These findings fully validate the critical role of calcium-dependent proteases in synaptic damage.

Beyond protease activation, Ca²⁺ influx also drives ROS production through dual pathways. Excessive ROS further impairs synaptic function

via oxidative stress (Bao et al., 2018). On one hand, elevated intracellular Ca^{2+} concentrations activate neuronal nitric oxide synthase (nNOS), prompting robust nitric oxide (NO) generation (Weaver et al., 2002). On the other hand, excessive Ca^{2+} influx into mitochondria induces mitochondrial Ca^{2+} overload, which in turn inhibits mitochondrial respiratory complex activity and substantially boosts ROS production (Adam-Vizi and Starkov, 2010). The synergistic effect of these pathways is particularly evident in the substantia nigra of PD mice: nNOS activity triples, leading to a twofold increase in NO levels (Siow et al., 2013); mitochondrial respiratory complex I activity is notably reduced, concurrently triggering a significant surge in ROS generation (Paradies et al., 2002). This excessive ROS initiates oxidative assaults on synaptic proteins such as receptors and transporters, as well as lipid membranes, both compromising synaptic structural integrity and disrupting signal transmission capabilities.

Furthermore, Ca^{2+} overload-induced mitochondrial dysfunction constitutes an ROS-independent toxic pathway that further exacerbates synaptic impairment. Mitochondria act not only as cellular "powerhouses" but also as critical hubs for intracellular Ca^{2+} storage and buffering; their stable function is indispensable for synaptic maintenance. When mitochondria sustain damage from Ca^{2+} overload, the initial change involves reduced mitochondrial membrane potential, followed by aberrant opening of the mitochondrial permeability transition pore (mPTP). Upon mPTP opening, mitochondrial apoptotic factors such as cytochrome C are released into the cytoplasm, directly initiating the neuronal apoptotic cascade (Thomas et al., 2012). This damaging process is clearly observable in spinal motor neurons from ALS mice: mitochondrial membrane potential is markedly diminished, mPTP opening rate is significantly increased, and cytochrome C release is substantially enhanced. Ultimately, this not only exacerbates post-synaptic neuronal apoptosis but also further impairs synaptic function due to disrupted energy supply (Pedrini et al., 2010).

3.4. Synaptic mitochondrial dysfunction

As the core structure for neuronal signal transmission, synapses rely on a continuous supply of adenosine triphosphate (ATP) for physiological processes including neurotransmitter release and ion gradient maintenance, with synaptic mitochondria acting as the primary source of this energy (Chen and Chan, 2009; Reddy and Beal, 2005). A long-term and systematic research conducted by Dr. Reddy's team has confirmed that synaptic mitochondria perform dual functions in energy metabolism and signal regulation (John and Reddy, 2021; Manczak et al., 2006; Pradeepkiran et al., 2020; Reddy et al., 2012). They generate the vast majority of synaptic ATP through oxidative phosphorylation to fuel synaptic activities. Meanwhile, they regulate calcium homeostasis, participate in the synthesis of proteins associated with synaptic plasticity, and maintain the structural integrity of synapses. In addition, the axonal transport of synaptic mitochondria and the balanced dynamics of mitochondrial fusion and fission, which are regulated by fusion proteins such as Mitofusin 1/2 (MFN1/2) and optic atrophy 1 (OPA1) as well as fission proteins such as dynamin-related protein 1 (Drp1), directly affect their functional performance and serve as an important guarantee for normal synaptic physiological activities (John and Reddy, 2021; Pradeepkiran et al., 2020; Reddy et al., 2012).

In neurodegenerative diseases, synaptic mitochondrial dysfunction often occurs at an early stage and drives disease progression. The collapse of synaptic mitochondrial function is already detectable in the early phase of AD (Reddy, 2014). Studies on post-mortem brain tissues from AD patients and APP transgenic mouse models have shown that amyloid-beta oligomers form complexes with Drp1, which significantly enhances the fission activity of Drp1, induces the abnormal accumulation of fragmented mitochondria, and then inhibits the expression of MFN2. These changes lead to a decline in oxidative phosphorylation efficiency and a substantial reduction in synaptic ATP production

(Manczak and Reddy, 2012; Nakamura and Lipton, 2010; Reddy and Beal, 2008). At the same time, phosphorylated tau binds to Drp1 and exacerbates mitophagy impairment, preventing the clearance of damaged mitochondria. This process in turn causes the degradation of key proteins such as synaptophysin and PSD95, and ultimately results in synaptic loss and cognitive deficits (Morton et al., 2021; Reddy and Oliver, 2019). The small-molecule inhibitor DDQ can restore considerable synaptic mitochondrial function by blocking the interaction between amyloid-beta and Drp1, which verifies the pathogenic role of this pathway (Kuruva et al., 2017). In HD, studies on neurons differentiated from patient-derived induced pluripotent stem cells and transgenic mouse models have demonstrated that mHTT oligomers directly target synaptic mitochondria. They enhance GTPase Drp1 enzymatic activity, a key factor in mitochondrial biogenesis, thereby reducing the generation of synaptic mitochondria. They also disrupt the axonal transport mechanism, leading to a decrease in mitochondrial density in synaptic regions (Reddy and Shirendeb, 2012). Electron microscopy observations have revealed that the cristae structure of synaptic mitochondria in HD patients is severely disrupted, with the activity of respiratory chain complex I reduced. Impaired ATP synthesis triggers the failure of synaptic transmission in medium spiny neurons of the striatum, which is closely related to the typical motor dysfunction phenotype of HD (Shirendeb et al., 2011). The hypothesis of mitochondrial dynamic imbalance and mitophagy impairment in PD has been confirmed by multiple studies (Krzystek et al., 2021). In the synapses of dopaminergic neurons in the substantia nigra of PD patients, the dysfunction of the PTEN-induced kinase 1 (PINK1)/parkin RBR E3 ubiquitin ligase (Parkin) pathway leads to impaired clearance of mitochondria through mitophagy and the abnormal accumulation of fragmented mitochondria. These damaged mitochondria show a significant decrease in respiratory chain complex I activity, which not only reduces ATP production but also releases reactive oxygen species to induce the aggregation of alpha-synuclein. The aggregated alpha-synuclein further inhibits mitochondrial transport, thus forming a vicious cycle. This mechanism shares high similarity with the impaired clearance of damaged mitochondria in AD, suggesting that mitophagy impairment may represent a common pathological link across neurodegenerative diseases (Martinez et al., 2018). The pathogenesis of ALS is associated with abnormal regulatory interactions between mitochondria and the endoplasmic reticulum (Reddy, 2008; Yipeng et al., 2025). In the synapses of motor neurons from ALS patients, the structure of mitochondria-associated endoplasmic reticulum membranes is disrupted. This defect prevents pyruvate, an intermediate product of glucose metabolism, from entering mitochondria for oxidative energy production, forcing cells to switch to fatty acid metabolism. This metabolic shift reduces the oxidative phosphorylation efficiency of synaptic mitochondria, resulting in insufficient ATP production. It also causes disordered electron flow in the respiratory chain, with a selective decrease in complex I activity that is particularly prominent in the spinal cord tissue. These changes ultimately lead to the degeneration of motor synapses and drive the progression of ALS (Chen and Chan, 2009; Larrea et al., 2025).

In summary, synaptic mitochondrial dysfunction plays a central role in the pathological processes of multiple neurodegenerative diseases. The underlying mechanisms vary with different disease types but all revolve around the decline in oxidative phosphorylation efficiency, insufficient ATP production, and the destruction of synaptic structure and function. These findings provide an important theoretical basis for the targeted therapy of such diseases.

3.5. Oxidative stress-mediated synaptic damage

Synapses serve as the primary loci for ROS production. ROS are generated via the mitochondrial respiratory chain, NMDAR activation, and the action of monoamine oxidases (Kalinichenko et al., 2023). Under physiological conditions, cells sustain ROS homeostasis through antioxidant enzymes such as SOD and GSH-Px. In neurodegenerative

diseases, however, this equilibrium is perturbed, oxidative stress is amplified, and synapses are impaired through multiple interconnected pathways.

ROS directly modulates the function of key synaptic proteins by inducing oxidative modifications at specific residues, thereby disrupting the structural foundation of synapses and compromising signal transmission efficacy (Beckhauser et al., 2016). This damage pattern is particularly pronounced in AD and PD: Notably, in AD, A β O-induced ROS specifically oxidizes the Cys10 residue of the presynaptic protein SYN. Oxidation at this site diminishes SYN's capacity to bind synaptic vesicles, directly attenuating neurotransmitter release efficiency (Lipton, 2022). In PD specifically, α -synO-induced ROS that targets and oxidizes tyrosine hydroxylase (TH) localized to the presynaptic membrane of dopaminergic neurons (Deas et al., 2016). As a key rate-limiting enzyme in dopamine biosynthesis, TH inactivation by ROS reduces dopamine production, ultimately impairing the signal transmission function of dopaminergic synapses.

Beyond protein modification, ROS also damages the lipid composition of synaptic membranes and interferes with the precision of synaptic signaling by altering membrane physical properties. Synaptic membranes are primarily composed of phospholipids, among which polyunsaturated fatty acids (PUFAs) are highly susceptible to ROS-induced oxidation due to their chemical structure—a process that generates oxidative byproducts such as malondialdehyde (MDA) (Murphy, 1985). MDA further undergoes cross-linking reactions with membrane lipids and proteins, leading to a marked reduction in synaptic membrane fluidity (Cheng et al., 2011). The consequences of this lipid oxidative damage are evident in striatal synapses of HD mice: MDA levels are notably elevated, accompanied by a significant decline in synaptic membrane fluidity (Pérez-Severiano et al., 2000). Membrane fluidity is indispensable for the proper localization and trafficking of neurotransmitter receptors such as AMPAR and NMDAR, as well as ion channels; its reduction directly hinders the speed and accuracy of synaptic signal transmission.

Furthermore, ROS-induced synaptic damage extends to the cellular antioxidant defense system. By suppressing the activity of antioxidant enzymes, ROS disrupt the balance between oxidative stress and antioxidant capacity, leading to progressive exacerbation of synaptic damage (Ahmad et al., 2017). This mechanism is most prominent in ALS. The activity of mutant SOD1 in ALS patients is already substantially diminished compared to healthy individuals. Excessive ROS further oxidatively modifies wild-type SOD1 in cells, damaging its active site and abrogating its antioxidant function (Rosen et al., 1994). Under the combined effect of these two factors, the overall cellular antioxidant capacity continues to decline, failing to effectively scavenge continuously generated ROS. These result in sustained ROS accumulation in synaptic regions, with a corresponding gradual aggravation of synaptic impairment.

3.6. Inflammatory response-mediated synaptic damage

Microglia and astrocytes serve as the primary immune effector cells of the CNS. Under physiological conditions, they contribute to the maintenance of synaptic homeostasis (Pinto et al., 2024); In neurodegenerative diseases, however, both cell populations undergo aberrant activation and drive synaptic impairment via releasing pro-inflammatory cytokines, activating the complement system, and directly phagocytosing synapses.

Known as the “immune sentinels” of the CNS, microglia secrete substantial levels of pro-inflammatory cytokines (e.g., TNF- α , IL-1 β , IL-6) following aberrant activation, representing the central initiating event of inflammatory synaptic damage (Marinelli et al., 2019). Across different diseases, these pro-inflammatory cytokines exert destructive effects by targeting distinct synaptic molecules. In AD, TNF- α released by A β O-activated microglia binds to tumor necrosis factor receptor 1 (TNFR1) at the postsynaptic membrane. This binding triggers

intracellular signaling cascades that promote AMPAR endocytosis, directly reducing the receptor density on the postsynaptic membrane (Steeland et al., 2018). In PD, IL-1 β secreted by α -synO-activated microglia suppresses DA release from the presynaptic membrane of dopaminergic synapses, impairing signal propagation in dopaminergic neural circuits (Pike et al., 2021). Corresponding experimental evidence directly validates this damaging effect: TNF- α expression in the hippocampus of APP/PS1 mice is markedly elevated, coinciding with a significant increase in AMPAR endocytosis (Steeland et al., 2018). Beyond pro-inflammatory cytokines, aberrant complement system activation constitutes another key pathway of inflammation-mediated synaptic damage. Its core mechanism involves the clearance of functional synapses via a “label-phagocytosis” mechanism. Upon activation, microglia secrete complement components, including C1q and C3. Among these, C1q acts as a “recognition signal” that binds to specific synaptic surface proteins, labeling normal synapses as “targets for clearance” (Dejanovic et al., 2022). Subsequently, microglia recognize these labeled synapses through complement receptor 3 (CR3) on their surface and eliminate them via phagocytosis (Stevens et al., 2007). This excessive “misdirected clearance” is observed in multiple diseases: in the brains of AD patients, hippocampal colocalization of C1q and SYN is significantly increased, accompanied by enhanced C3 deposition (Gomez-Arboledas et al., 2021); and the number of microglia engaged in synaptic phagocytosis is doubled. In the spinal cord of ALS mice, C1q expression is notably upregulated, and the proportion of motor neuron synapses undergoing phagocytosis is significantly elevated. Excessive complement system activation results in misdirected clearance of numerous functionally intact synapses, directly exacerbating neurological dysfunction (Clement et al., 2003).

Of note, aberrantly activated astrocytes deviate from their role as “neuroprotective regulators” and contribute to synaptic damage through multiple pathways. In addition to downregulating the glutamate transporter GLT-1 as previously described to impair glutamate clearance, astrocytes also secrete toxic factors such as complement C3 and TNF- α that act directly on synapses (Luchena et al., 2018). To validate this mechanism, researchers established a co-culture system of ALS patient-derived astrocytes and motor neurons. Results demonstrated that astrocyte-secreted C3 binds to CR3 at the postsynaptic membrane of motor neurons, compromising the structural integrity of the postsynaptic membrane and ultimately leading to a significant reduction in synaptic transmission efficacy. This experiment further confirms the critical role of astrocytes in inflammation-associated synaptic damage (Taga et al., 2019).

3.7. Neurotrophic factor deficiency

Neurotrophic factors (e.g., BDNF, NGF, GDNF) serve as critical signaling molecules that sustain synaptic structural integrity and functional homeostasis. They regulate synaptic protein biosynthesis, synaptic plasticity, and neuronal survival (Vecchiarelli et al., 2024). In neurodegenerative diseases, aberrations in neurotrophic factor expression or their signaling cascades deprive synapses of trophic support, culminating in synaptic impairment.

As a core trophic factor governing synaptic survival and plasticity, BDNF downregulation represents a shared hallmark across multiple neurodegenerative diseases, with its regulatory mechanisms tightly linked to disease-specific pathogenic proteins (S et al., 2025). In AD, A β O suppresses the activity of cAMP response element-binding protein (CREB), directly diminishing BDNF transcription (Vitolo et al., 2002). In PD, α -synO primarily targets substantia nigra dopaminergic neurons, inhibiting BDNF synthesis and secretion (Ma et al., 2025). Corresponding experimental evidence clearly validates this aberration: BDNF mRNA transcription is markedly diminished in the hippocampus of APP/PS1 mice, accompanied by a substantial reduction in protein levels (Ma et al., 2025); while BDNF secretion is notably impaired in the substantia nigra of A53T mice. BDNF deficiency directly undermines its

trophic support to synapses, compromising synaptic survival, stability, and plasticity regulation (Kohno et al., 2004). More importantly, even in the presence of residual BDNF, impairment of its downstream signaling cascades further exacerbates synaptic damage, precluding synapses from receiving trophic signals normally. Under physiological conditions, BDNF must bind to tyrosine kinase B (TrkB) receptors to activate downstream pathways including PI3K/Akt and MAPK/ERK, thereby regulating synaptic protein biosynthesis and LTP induction (Schildt et al., 2013). In pathological states, however, this signal transduction process is disrupted: in AD, p-tau directly interacts with TrkB receptors, inhibiting receptor dimerization and phosphorylation and abrogating signal transduction at the initiation stage (Chen et al., 2021); in HD, mHTT binds to key molecules in the ERK pathway, blocking downstream signal propagation (Bodai and Marsh, 2012).

Beyond BDNF, GDNF deficiency exhibits selective implication in dopaminergic synaptic impairment in PD, representing one of the key drivers of dopaminergic synaptic dysfunction (Kim and Chang, 2024). GDNF is a trophic factor specific to dopaminergic synapses, whose core function is to promote dopaminergic neuronal survival and synaptic repair—both indispensable for maintaining the stability of dopaminergic

neural circuits (Chen et al., 2025). In the substantia nigra of PD patients, GDNF expression secreted by astrocytes is notably reduced, directly impairing the intrinsic repair capacity of DAergic synapses (Virachit et al., 2019). Complementary experiments involving GDNF supplementation have confirmed its protective role: in the 6-OHDA-induced PD rat model, exogenous GDNF administration notably increases striatal dopaminergic synaptic density and significantly enhances DA release (Smith et al., 2005). This fully demonstrates GDNF's irreplaceable role in dopaminergic synaptic survival and functional recovery.

To conclude, synaptic damage in neurodegenerative diseases is initiated by pathological protein aggregation, exacerbated via dysregulated calcium homeostasis, propelled by oxidative stress and neuroinflammation, and ultimately culminates in functional deficits arising from synaptic depletion (Fig. 3). Elucidation of this pathological cascade not only furnishes a “core regulatory paradigm” for interpreting the disease-specific pathological features but also establishes a pivotal theoretical underpinning for developing cross-disease universal synaptic protection strategies.

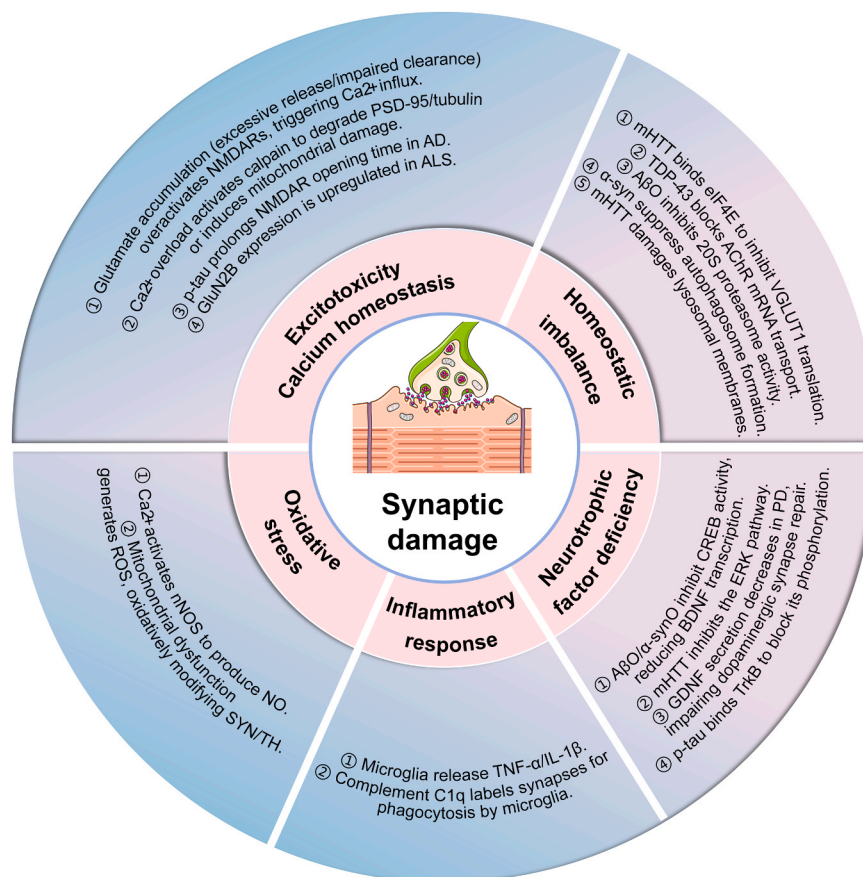


Fig. 3. Multidimensional pathological mechanisms of synaptic damage in neurodegenerative diseases. Excitotoxicity/Calcium homeostasis: Excessive glutamate release or impaired clearance, accompanied by abnormal activation of NMDAR/AMPA, induces calcium influx overload; meanwhile, excessive Cdk5 activation and degradation of PSD-95 and tubulin contribute to synaptic structural and functional damage in AD and ALS, respectively. Homeostatic imbalance: mHTT binds to molecules such as TOP and impairs DPP6, interfering with synaptic function regulators like CBP and FMRP. This disrupts synapse-related molecular homeostasis and drives the process of synaptic damage. Neuron damage: α-syn inhibits CREB activity; mHTT interferes with BDNF and ERK pathways; p-tau binds to Tau and blocks its phosphorylation. These multi-target effects on neuronal signaling and structure directly impair synaptic function. Inflammatory response/Oxidative stress: Microglia release inflammatory cytokines such as $\text{TNF-}\alpha$ and $\text{IL-1}\beta$, and complement C1q labels synapses to mediate their phagocytosis by microglia; meanwhile, iNOS produces NO, and Cdk5 activates pathways like Nox to trigger oxidative stress, which synergistically exacerbates synaptic damage. NMDAR (N-Methyl-D-aspartate receptor), AMPAR (alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor), Cdk5 (cyclin-dependent kinase 5), PSD-95 (postsynaptic density protein 95), AD (alzheimer's disease), ALS (amyotrophic lateral sclerosis), mHTT (mutant huntingtin protein), TOP (terminal oligopyrimidine tract-binding protein), CBP (CREB-binding protein), FMRP (fragile X mental retardation protein), CREB (cAMP response element-binding protein), BDNF (brain-derived neurotrophic factor), ERK (extracellular signal-regulated kinase), p-tau (phosphorylated tau), $\text{TNF-}\alpha$ (tumor necrosis factor-alpha), $\text{IL-1}\beta$ (interleukin-1 beta), iNOS (inducible nitric oxide synthase), Nox (NADPH oxidase). Figure created with PowerPoint.

4. Molecular pathway mechanisms of synaptic damage in major neurodegenerative diseases

4.1. Abnormal aggregation of pathogenic proteins destroys the synaptic molecular basis

Synaptic impairment in the four major neurodegenerative disorders (AD, PD, HD, ALS) shares a common initial driver: the abnormal aggregation of disease-specific pathogenic proteins. While these pathogenic proteins (A β /p-tau in AD, α -synuclein in PD, mHTT in HD, TDP-43/SOD1 in ALS) vary in molecular structure, they uniformly disrupt the homeostatic balance of the presynaptic membrane, postsynaptic membrane, and synaptic cleft. Two core mechanisms underpin this disruption: direct binding to key synaptic molecules or interference with synaptic material transport, which sets the stage for subsequent cascading damage.

Pathogenic proteins directly target core functional molecules at the presynaptic membrane, inhibiting normal neurotransmitter release. In AD, A β oligomers interact with the PrPC/Fyn complex, overactivating Cav2.1 calcium channels and causing dysregulated Ca²⁺ influx and imbalanced synaptic vesicle release (Jeans et al., 2025). In PD, α -synuclein oligomers bind to the SNARE complex, hindering synaptic vesicle-membrane fusion (Yoo et al., 2021). In HD, mHTT associates with HAP1, blocking vesicle microtubule transport and reducing glutamate supply (Li et al., 2003). In ALS, aggregated mutant SOD1/TDP-43 impairs the stability of VGLUT1, leading to abnormal glutamate release (Broadhead et al., 2023). The signal transduction function of the postsynaptic membrane is also disrupted by pathogenic proteins. In AD, p-tau oligomers bind to PSD-95/Shank3, destabilizing NMDARs/AMPA receptors and inducing their endocytosis (Shen et al., 2023). PD, α -synuclein oligomers activate calpain proteases to degrade PSD-95 (Shams et al., 2019). In HD, nuclear translocation of mHTT inhibits DARPP-32 transcription, reducing the sensitivity of dopamine receptors (Fjodorova et al., 2019). In ALS, TDP-43 aggregation reduces the transport of AChR mRNA, resulting in insufficient receptor clustering at the postsynaptic membrane of the NMJ (Bjork et al., 2022). The homeostatic balance of the synaptic cleft microenvironment is also disrupted by pathogenic proteins. In AD, A β oligomers inhibit the activity of acetylcholinesterase, slowing neurotransmitter degradation (Alvarez et al., 1998). In ALS, downregulation of GLT-1 in astrocytes causes impaired glutamate clearance (Vanoni et al., 2004). More importantly, the pathogenic proteins of all four diseases induce decreased expression of adhesion molecules such as N-cadherin, directly undermining the structural stability of synapses.

4.2. Calcium homeostasis dysregulation drives the synergistic amplification of oxidative stress and inflammation

In the synaptic damage process of the four major neurodegenerative diseases, dysregulated calcium homeostasis acts as a central intermediate link bridging the earlier and later stages. Ultimately, the destruction of the synaptic molecular basis by pathogenic proteins triggers an abnormal increase in local Ca²⁺ concentration at synapses. Excessive Ca²⁺ further activates two major toxic pathways: oxidative stress and inflammatory response. This forms a vicious cycle where calcium overload induces oxidative stress and inflammation, which in turn promotes more severe calcium overload. This cycle continuously accelerates synaptic damage.

While the sources of Ca²⁺ overload vary among the four diseases, they essentially fall into two categories: excessive Ca²⁺ influx or impaired Ca²⁺ clearance. In AD, A β oligomers activate NMDARs and Cav2.1 channels (Jeans et al., 2025); In ALS, excessive glutamate accumulation overactivates NMDARs (Foran and Trotti, 2009). Both cases are dominated by increased exogenous Ca²⁺ influx. In PD, α -synuclein oligomers disrupt endoplasmic reticulum IP3 receptors and mitochondrial MCU (Ramezani et al., 2023). In HD, mHTT causes

abnormal calcium channel activity (Miranda et al., 2019). These cases are mainly characterized by impaired intracellular Ca²⁺ storage and clearance. Regardless of the trigger, the final result is a marked increase in free Ca²⁺ concentration in synaptic regions. For example, Ca²⁺ influx in the hippocampus of AD mice is significantly elevated (Yeung et al., 2021), and the Ca²⁺ concentration in the substantia nigra of PD mice is notably heightened (Porras et al., 2012).

Excessive Ca²⁺ first induces oxidative stress through two pathways. On one hand, it activates neuronal nitric oxide synthase (nNOS). For instance, nNOS activity in the substantia nigra of PD mice is significantly increased, and NO production is also markedly elevated (Siow et al., 2013); On the other hand, excessive Ca²⁺ enters mitochondria, causing mitochondrial calcium overload and inhibiting the activity of mitochondrial respiratory chain complexes. For example, the activity of complex I in the hippocampus of AD mice is notably reduced (Paradies et al., 2002). Ultimately, these changes lead to a significant increase in ROS production. These ROS oxidatively modify synaptic proteins (e.g., the Cys10 site of SYN in AD) and membrane lipids (e.g., MDA content in the striatum of HD mice is markedly elevated), further exacerbating synaptic damage.

Meanwhile, Ca²⁺ overload also activates glial cells and triggers inflammation. In AD, Ca²⁺ overload induced by A β oligomers activates the NLRP3 inflammasome in microglia, leading to increased IL-1 β levels in the hippocampus (Ahmed et al., 2017); In PD, α -synuclein oligomers promote microglial expression of complement C3 through Ca²⁺-dependent pathways, resulting in notable C3 deposition in the substantia nigra (Stevens et al., 2007); In HD and ALS, calcium overload causes neurons to release damage-associated molecular patterns (DAMPs), which recruit microglia to phagocytose synapses (Kawamata et al., 2014). The synaptic phagocytosis rate in the spinal cord of ALS mice is significantly increased. Additionally, under calcium overload, astrocytes downregulate GLT-1 (GLT-1 expression in the spinal cord of ALS mice is notably reduced), which further exacerbates glutamate accumulation and calcium overload, strengthening the vicious cycle (Pardo et al., 2006).

4.3. Common manifestations of synaptic structure loss and functional abnormalities

After cascading reactions such as pathogenic protein aggregation and dysregulated calcium homeostasis, the four major neurodegenerative diseases eventually exhibit shared terminal features: reduced synaptic count and impaired synaptic function. This type of damage occurs earlier than neuronal soma death and is directly associated with clinical symptoms such as cognitive impairment and motor abnormalities.

Although synaptic structural damage is concentrated in different brain regions, it shares consistent characteristics, all manifested as decreased dendritic spine density and abnormalities of the presynaptic and postsynaptic membranes. For example, the dendritic spine density in the hippocampus of AD patients is significantly reduced (Scheff et al., 1990). The number of dopaminergic synapses in the striatum of PD patients is notably lowered (Gordán-Vélez et al., 2021). All four diseases are accompanied by disruption of the active zone on the presynaptic membrane (e.g., decreased expression of SYN in AD) and damage to the PSD structure on the postsynaptic membrane (e.g., decreased expression of Shank3 in HD).

Synaptic functional abnormalities are reflected in reduced transmission efficiency and impaired plasticity. In terms of transmission, the amplitude of excitatory postsynaptic potential (EPSP) in the hippocampus of AD patients is notably decreased (Gelman et al., 2018), and DA release in the striatum of PD patients is significantly reduced (Gordán-Vélez et al., 2021). In terms of plasticity, all four diseases exhibit difficulty in LTP induction and abnormally enhanced LTD. For instance, the amplitude of LTP in the hippocampus of AD mice is substantially reduced, ultimately leading to impairments in learning and memory or motor control (Shankar et al., 2008).

Taken together, although synaptic damage in the four major neurodegenerative diseases involves disease-specific details (Table 2), all adhere to a unified common pathway: damage initiation via pathogenic protein aggregation, toxicity amplification through dysregulated calcium homeostasis, process acceleration by oxidative stress and inflammation, and eventual functional impairment due to synaptic loss. The clarification of this pathway not only provides a core framework of rules for understanding the pathological mechanisms of different diseases but also lays a key theoretical foundation for the development of cross-disease universal synaptic protection strategies, such as targeting calcium homeostasis regulation and anti-inflammatory therapy.

5. Early diagnostic biomarkers associated with synaptic damage

Synaptic damage manifests prior to neuronal soma death and the onset of clinical symptoms, with its associated molecular signatures holding promise as potential biomarkers for the early diagnosis of neurodegenerative diseases. Current research centers on CSF and blood-based molecular biomarkers (Table 3).

5.1. CSF biomarkers

5.1.1. Synaptic structural protein biomarkers

Distinct synaptic structural proteins offer multi-dimensional insights into synaptic membrane integrity based on their unique subcellular localization and functional roles. Among these, alterations in CSF levels of presynaptic membrane-specific proteins and postsynaptic membrane-associated proteins represent the most extensively investigated biomarker classes.

SYN is a presynaptic membrane-specific protein. Fluctuations in its CSF concentration directly correlate with presynaptic membrane

integrity, and reduced levels serve as a direct indicator of presynaptic impairment (Olczak et al., 2025). A clinical study demonstrated that CSF SYN concentrations were lower in AD patients compared to healthy controls, with this downward trend negatively correlating with hippocampal volume loss (Sze et al., 1997). Additionally, this protein holds value for the early diagnosis of PD: CSF SYN levels are decreased in patients with early-stage PD, and this reduction correlates with diminished striatal DAT density (Soni and Shah, 2025). Thus, it can serve as a marker to assess early dopaminergic synaptic damage in PD. In contrast to SYN, PSD-95 is a key postsynaptic scaffolding protein. Alterations in its CSF concentration mirror the structural stability of the postsynaptic membrane, with elevated levels typically signifying postsynaptic impairment. In AD patients, CSF PSD-95 concentrations are higher than in controls, and this level positively correlates with reduced MMSE scores—indicating that higher PSD-95 levels correspond to more severe cognitive dysfunction (Kivisäkk et al., 2022). In patients with preclinical HD, CSF PSD-95 concentrations increase prematurely, occurring 1–2 years before the onset of clinical symptoms. This makes it a crucial early warning indicator for incipient synaptic damage in HD (Fourie et al., 2014).

Beyond scaffolding proteins, postsynaptic calcium regulatory proteins also contribute to synaptic damage assessment, with Neurogranin standing as a paradigmatic example. Predominantly expressed in cortical and hippocampal neurons, it regulates postsynaptic calcium signaling (Neuner-Jehle et al., 1996). CSF Neurogranin concentrations are significantly elevated in AD patients. More importantly, it exhibits robust diagnostic sensitivity and specificity for AD (Agnello et al., 2021). This diagnostic efficacy outperforms the conventional Aβ42/p-tau ratio, enabling effective differentiation of AD patients from healthy individuals and distinguishing AD from other neurodegenerative disorders. This provides a novel direction for the precise diagnosis of AD.

Table 2
Comparison of synaptic damage characteristics in major neurodegenerative diseases.

Disease name	Key sites of synaptic damage	Critical pathological proteins	Major manifestations of synaptic damage	Ref.
AD	Early: Hippocampus (CA1 region), entorhinal cortex; Late: Prefrontal cortex, temporal cortex	AβO, p-tau	1. Decreased synaptic density, which is negatively correlated with MMSE scores. 2. Dendritic spine atrophy and decreased quantity. 3. Disrupted presynaptic active zone and reduced synaptic vesicles; 4. AβO inhibits LTP and enhances LTD abnormally.	(Bose et al., 2024; Rolland et al., 2020; Scheff et al., 1990; Terry et al., 1991)
PD	Nigrostriatal dopaminergic synapses, locus coeruleus noradrenergic synapses, raphe nucleus serotonergic synapses	α-synO, LBS	1. Reduced striatal dopaminergic synapses and decreased DA content. 2. α-synO inhibits SNARE complex assembly, reducing DA release. 3. Downregulated D1/D2 receptor expression on striatal MSNs and weakened binding to PSD–95.	(Ba and Martin, 2015; Brolin et al., 2023; Gordián-Vélez et al., 2021; Mehra et al., 2019; Porras et al., 2012)
HD	Striatum (corticostriatal glutamatergic synapses, striatopallidal GABAergic synapses)	mHTT	1. Decreased density of dendritic spines in striatal neurons. 2. mHTT binds HAP1 to block vesicular transport, reducing glutamate/GABA release. 3. Decreased calcium channel activity and reduced Ca ²⁺ influx. 4. Difficulty in LTP induction and abnormally enhanced LTD in the striatum.	(Li et al., 2003; Silva et al., 2017; Spires et al., 2004; Twelvetrees et al., 2010; Usdin et al., 1999)
ALS	NMJs, corticospinal motor synapses	Mutant SOD1, TDP–43, FUS	1. Reduced presynaptic vesicles, shallow postsynaptic folds, and decreased AChR density at NMJs. 2. Reduced corticospinal synapses and abnormal glutamate release causing excitotoxicity. 3. Downregulated GLT–1 in astrocytes, impairing glutamate clearance. 4. Microglial release of inflammatory factors damaging synapses.	(Fogarty et al., 2015; McIntosh et al., 2023; Raghavendra Rao et al., 2000; Rao et al., 2012)

Table 3

Early diagnostic biomarkers associated with synaptic damage.

Biomarker category	Specific biomarkers	Corresponding disease	Diagnostic value (trend/diagnostic efficacy)	Ref.
CSF biomarkers	SYN	AD/PD	Decreased in AD (negatively correlated with hippocampal volume); decreased in early PD (correlated with striatal DAT density).	(Soni and Shah, 2025; Sze et al., 1997)
	PSD-95	AD/HD	Increased in AD (negatively correlated with MMSE scores); increased in early HD (1–2 years prior to clinical symptoms).	(Fourie et al., 2014; Kivisäkk et al., 2022)
	Neurogranin	AD	Increased; diagnostic sensitivity and specificity for AD (superior to A β 42/p-tau ratio)	(Agnello et al., 2021)
	A β O	AD	Increased in preclinical AD (negatively correlated with hippocampal synaptic density; used for ultra-early screening).	(Hölttä et al., 2013)
	p-tau oligomers	AD/PD	Increased in AD (sensitivity, specificity); increased in PD (aids in differentiating movement disorders).	(Blömeke et al., 2022; Tai et al., 2012)
Blood biomarkers	SYN-F	AD/PD	Decreased in AD; decreased in PD.	(Meng et al., 2022; Parnetti et al., 2019)
	PSD-F	AD/HD	Increased in AD (positively correlated with CSF PSD-95); decreased in HD (earlier than imaging abnormalities).	(Kivisäkk et al., 2022; Zarate et al., 2021)
	Exosome-encapsulated A β O/ α -synO	AD/PD	A β O increased in AD exosomes; α -synO increased and GDNF decreased in PD exosomes.	(Goetzl et al., 2016; Zheng et al., 2021)

5.1.2. Synaptic function-related molecular biomarkers

Among molecular biomarkers associated with synaptic function, glutamate metabolites and BDNF offer insights into synaptic functional abnormalities through two core dimensions: metabolic balance, and trophic support. Alterations in their CSF levels furnish critical evidence for assessing synaptic impairment across distinct neurodegenerative diseases.

Glutamate metabolite levels directly correlate with synaptic excitotoxicity. Elevated CSF levels of these metabolites frequently signify synaptic dysfunction—a characteristic consistently observed in both ALS and AD. In ALS patients, CSF glutamate levels are higher relative to healthy controls, with these levels positively correlating with disease progression rate, implying that aberrant glutamate metabolism may accelerate disease advancement (Spreux-Varoquaux et al., 2002). In AD patients, CSF glutamate levels exhibit a modest elevation, yet this change is tightly linked to the extent of hippocampal synaptic loss. It can indirectly reflect the severity of synaptic damage, providing a reference for evaluating synaptic dysfunction in AD (Madeira et al., 2018). In contrast to glutamate metabolites that mediate “functional regulation”, BDNF acts as a pivotal trophic factor sustaining synaptic survival and function. Reduced BDNF levels in CSF typically indicate insufficient neurotrophic support, which may induce or exacerbate synaptic impairment (Li et al., 2009). In PD patients, BDNF concentrations in CSF are diminished, and the extent of this reduction correlates with the severity of dopaminergic synaptic impairment in the substantia nigra. Thus, it serves as a key indicator for assessing dopaminergic synaptic damage in PD (Nagatsu et al., 2000). In AD patients, BDNF levels in CSF are also decreased. More importantly, tracking BDNF level fluctuations can predict the rate of cognitive deterioration in patients, furnishing a reliable basis for dynamic monitoring of AD progression (Diniz et al., 2014).

5.1.3. Pathogenic protein derivative biomarkers

Among pathogenic protein-derived biomarkers linked to synaptic damage, A β O and p-tau oligomers emerge as pivotal indicators reflecting synaptic impairment in neurodegenerative diseases, owing to their direct synaptic toxicity. Alterations in their CSF levels furnish critical evidence for the ultra-early detection of AD, as well as the assessment of shared and disease-specific synaptic damage in AD and PD, respectively.

Soluble A β O acts as a direct pathogenic mediator of early synaptic impairment in AD (Esparza et al., 2013). The time point of its CSF level elevation precedes the formation of cerebral A β plaques by a substantial margin, a feature that positions it as a potential core indicator for AD's ultra-early diagnosis. Compelling research findings corroborate this notion: in patients with preclinical AD asymptomatic stage, CSF A β O levels are already markedly elevated compared to healthy controls, with

this change positively correlating with reduced hippocampal synaptic density—indicating that higher A β O levels correspond to more severe hippocampal synaptic damage. Thus, detecting CSF A β O levels enables ultra-early AD screening, securing a critical window for disease intervention (Hölttä et al., 2013). In contrast to A β O predominantly linked to AD, p-tau oligomers represent a shared biomarker implicated in synaptic damage across both AD and PD. However, their level fluctuations and clinical implications differ between the two conditions: In AD patients, CSF p-tau oligomer levels are notably higher than in healthy controls. The biomarker exhibits robust diagnostic sensitivity and specificity for AD, making it particularly suitable for screening preclinical AD asymptomatic stage (Tai et al., 2012). In PD patients, CSF p-tau oligomer levels show a modest elevation, yet this change directly correlates with dopaminergic synaptic impairment. It aids in differentiating PD from other analogous movement disorders, providing supplementary reference for PD's differential diagnosis (Blömeke et al., 2022).

5.2. Blood biomarkers

5.2.1. Synaptic protein fragments

Among blood-borne synaptic protein fragment biomarkers, synapsin fragments (SYN-F) and PSD-95 fragments (PSD-F) mirror the impairment status of the presynaptic and postsynaptic membranes, respectively. Since intact synaptic proteins struggle to traverse the blood-brain barrier (BBB), these degraded fragments emerge as critical surrogate indicators for central synaptic damage in peripheral blood, offering a viable approach for the early non-invasive assessment of diverse neurodegenerative diseases.

SYN-F primarily functions as a peripheral monitoring marker for presynaptic membrane impairment. Reduced plasma SYN-F levels directly correlate with structural and functional abnormalities of central presynaptic membranes—a relationship consistently demonstrated in both AD and PD. In AD patients, plasma SYN-F concentrations are lower than in healthy controls, facilitating the early detection of AD (Meng et al., 2022). In PD patients, plasma SYN-F levels exhibit a modest reduction, yet this change is strongly associated with striatal DAT imaging abnormalities. It indirectly reflects damage to dopaminergic presynaptic membranes, thus serving as a potential indicator for the early non-invasive detection of PD (Parnetti et al., 2019). In contrast to presynaptic-targeting SYN-F, fluctuations in plasma PSD-F concentrations mainly indicate the extent of postsynaptic membrane disruption and correlate with CSF biomarkers. In AD patients, plasma PSD-F levels are elevated compared to controls and positively correlate with CSF PSD-95 concentrations—enabling indirect inference of central postsynaptic membrane impairment via peripheral blood PSD-F detection (Kivisäkk et al., 2022). In HD patients, plasma PSD-F concentrations

decrease prematurely, preceding the onset of imaging abnormalities. This overcomes the constraint of “detection solely post-symptom onset” for early HD diagnosis, unlocking new possibilities (Zarate et al., 2021).

5.2.2. Synaptic biomarkers encapsulated in exosomes

In research on exosome-encapsulated synaptic biomarkers, exosomes—endowed with distinctive biological characteristics—act as a pivotal carrier bridging central synaptic impairment and peripheral blood-based detection, providing a superior alternative for the accurate diagnosis of neurodegenerative diseases. They not only ferry synaptic-

related proteins across the BBB seamlessly but also shield these proteins from degradation in the bloodstream. Compared to free molecular biomarkers, exosome-encapsulated biomarkers exhibit higher stability and specificity, effectively resolving the problem of inadequate detection sensitivity for synaptic damage indicators in peripheral blood (Hong et al., 2024).

This advantage is particularly pronounced in AD diagnosis. Studies have shown that in plasma exosomes from AD patients, neurotoxic soluble A β O levels are markedly elevated relative to healthy controls, while SYN concentrations are reduced. Furthermore, these molecular-level

Table 4
Targeted therapeutic strategies for synaptic protection.

Therapeutic direction	Specific interventions	Representative drugs/technologies	Efficacy (disease models/clinical settings)	Ref.
Inhibiting pathogenic protein toxicity	Reducing pathogenic protein production	γ -secretase inhibitor (semagacestat) BACE1 inhibitor (verubecestat) α -syn siRNA	Decreased A β O in the hippocampus of AD mice; increased synaptic density. Decreased A β 42 in the cerebrospinal fluid of AD patients; slowed decline rate of SYN. Decreased α -synuclein in the substantia nigra of PD mice; increased dopaminergic synaptic density.	(Saura, 2010) (Querol-Vilaseca et al., 2022) (Lewis et al., 2008)
	Inhibiting pathogenic protein aggregation	Anti-A β O monoclonal antibody (solanezumab)	Decreased A β O in the cerebrospinal fluid of AD patients; slowed decline rate of hippocampal synaptic density.	(Doody et al., 2014)
	Promoting pathogenic protein degradation	Anti- α -synO monoclonal antibody (prasinezumab)	Decreased α -synO in the substantia nigra of PD mice; reduced dopaminergic synapse loss.	(Pagano et al., 2024)
		Autophagy activator (rapamycin)	Increased A β degradation in the hippocampus of AD mice; increased α -synuclein degradation in the substantia nigra of PD mice.	(Cai et al., 2024; Liu et al., 2023)
Regulating synaptic protein homeostasis	Promoting synaptic protein synthesis	Proteasome activator (bortezomib)	Decreased p-tau in the hippocampus of AD mice; increased mHTT degradation in the striatum of HD mice.	(Dasgupta et al., 2015; Vulin et al., 2023)
		CREB activator (forskolin)	Increased CREB phosphorylation and BDNF in the hippocampus of AD mice; increased dopamine receptors in the striatum of PD mice.	(Alharbi et al., 2022; Rosa and Fahnestock, 2015)
	Improving protein folding	4E-BP1 inhibitor	Increased VGLUT1 translation in the striatum of HD mice; increased AChR translation in ALS mice.	(Peng et al., 2022; Talukder et al., 2021)
		HSP70 activator (geldanamycin)	Enhanced folding efficiency of PSD-95 in the hippocampus of AD mice; increased proportion of correctly folded α -synuclein in the substantia nigra of PD mice.	(Shen et al., 2005; Zare et al., 2015)
Alleviating excitotoxicity	Inhibiting excessive NMDAR activation	GluN2B subunit-selective antagonists (Ro25-6981, Ifenprodil) NMDAR partial agonists (D-cycloserine)	AD mice: reduced hippocampal NMDAR Ca ²⁺ influx, alleviated LTP; ALS mice: less NMDAR activity, mitigated Ca ²⁺ overload, improved NMJs density. PD mice show precisely regulated striatal MSN NMDAR activity, improved signal balance, and reduced motor scores.	(Li et al., 2011; Mirmotahari et al., 2024) (Rouaud and Billard, 2003)
	Enhancing glutamate clearance	GLT-1 activator (ceftriaxone)	Increased GLT-1 in the spinal cord of ALS mice; improved glutamate clearance rate; prolonged survival.	(Rothstein et al., 2005)
Regulating inflammatory response	Inhibiting microglial overactivation	Minocycline	Decreased activated microglia and increased SYN in the hippocampus of AD mice; reduced dopaminergic synapse loss in the substantia nigra of PD mice.	(Biscaro et al., 2012; Wu et al., 2002)
	Blocking complement activation	C1q-neutralizing antibody	Decreased C1q-synapse binding in the hippocampus of AD mice; increased synaptic density; restored striatal LTP in HD mice.	(Dejanovic et al., 2018; Wilton et al., 2023)
Supplementing neurotrophic factors	Exogenous supplementation	AAV-BDNF (hippocampal injection)	Increased BDNF in the hippocampus of AD mice; increased SYN and PSD-95 respectively.	(Jiao et al., 2016)
		AAV-GDNF (striatal injection)	Increased striatal DAT density in PD patients; improved motor scores.	(Van Laar et al., 2025)
	Endogenous regulation	TrkB agonist (7,8-Dihydroxyflavone)	Increased TrkB phosphorylation in the hippocampus of AD mice; restored LTP; reduced α -synuclein-induced damage in the substantia nigra of PD mice.	(Gao et al., 2016)
Mitophagy enhancer-mediated synaptic protection	Activating mitochondrial quality control pathways (PINK1/Parkin, autophagy receptors)	Urolithin A, nicotinamide riboside, PGC-1 α agonists	Improved damaged mitochondria clearance, synaptic ATP and connection in AD mice; restored dopaminergic synaptic ATP and neurotransmitter release in PD mice; enhanced striatal synaptic transmission and motor coordination in HD mice.	(Morton et al., 2021; Schöndorf et al., 2018; Tsunemi et al., 2012)
Novel strategies	Stem cell transplantation	MSC	Increased BDNF and IL-10 in the hippocampus of AD mice; increased synaptic density; prolonged survival of ALS mice.	(Guo et al., 2020; Terashima et al., 2020)
	Gut microbiota regulation	Probiotics (bifidobacterium/lactobacillus)	Increased gut short-chain fatty acids in AD mice (reduced hippocampal microglial activation); decreased α -synuclein aggregation in the substantia nigra of PD mice; increased dopaminergic synaptic density.	(Aktas et al., 2024; Zhu et al., 2021)

changes are strongly associated with the degree of hippocampal synaptic damage in patients, allowing concurrent assessment of AD pathological progression and synaptic impairment status (Goetzl et al., 2016). Beyond AD, exosome-encapsulated biomarkers also demonstrate substantial value in PD diagnosis. In plasma exosomes from PD patients, α -synO—a core PD pathological marker—is substantially elevated, directly signifying toxic damage to dopaminergic synapses. Meanwhile, concentrations of the neurotrophic factor GDNF are reduced, indicating inadequate neurotrophic support (Zheng et al., 2021). Simultaneous detection of these two molecules enables holistic evaluation of pathological protein toxicity and neurotrophic status in PD, further enhancing diagnostic accuracy and comprehensiveness.

6. Targeted synaptic protection therapeutic strategies

Given the central role of synaptic damage in neurodegenerative diseases, therapeutic approaches centered on synaptic protection have emerged as a prominent research focus, encompassing suppressing pathological protein toxicity, restoring synaptic protein homeostasis, alleviating excitotoxicity, modulating neuroinflammatory cascades, and administering neurotrophic factors (Table 4).

6.1. Inhibiting pathogenic protein toxicity

6.1.1. Reducing pathogenic protein production

In the context of A β biosynthesis in AD, current research centers on targeting pivotal enzymes in A β cleavage: γ -secretase and β -secretase (BACE1). Among these, γ -secretase inhibitors act by suppressing γ -secretase activity, thereby reducing the final cleavage of amyloid precursor protein (APP) into A β (Tagami et al., 2017). In the APP/PS1 transgenic mouse model that recapitulates AD pathology, semagacestat's therapeutic efficacy has been clearly validated: hippocampal A β O levels are decreased, while synaptic density is markedly elevated, successfully reversing synaptic impairment induced by excessive A β production (Saura, 2010).

Beyond γ -secretase, BACE1 also exhibits substantial therapeutic potential in AD via its inhibitors. Verubecestat, for instance, can selectively block APP's initial cleavage by BACE1, reducing the upstream substrates for A β biosynthesis at the source (Saura, 2010). In clinical trials involving AD patients, verubecestat's efficacy has been further confirmed: CSF levels of neurotoxic A β 42 are decreased, and more importantly, the decline rate of SYN is significantly slowed. This indicates the drug can effectively delay synaptic damage progression in AD patients, offering the potential to preserve cognitive function. In parallel with A β 's pathogenic mechanism in AD, a core pathological feature of PD is the excessive production and aberrant aggregation of α -syn. This process directly compromises the structure and function of dopaminergic synapses, making reducing α -syn production a key strategy for protecting dopaminergic synapses in PD patients. Current research in this area falls into two main categories: “transcriptional inhibition” and “translational regulation”. At the transcriptional level, small interfering RNA (siRNA) targeting the α -syn promoter can selectively bind to the α -syn gene's promoter region, inhibiting α -syn transcription and thus reducing its gene products (Lewis et al., 2008). In the S129D transgenic mouse model harboring the α -syn mutation, this siRNA treatment diminishes α -syn expression in the substantia nigra, effectively alleviating synaptic stress caused by excessive α -syn accumulation.

At the translational level, mammalian target of rapamycin (mTOR) inhibitors such as rapamycin indirectly reduce α -syn protein synthesis by regulating intracellular translational processes. Hyperactivation of the mTOR signaling pathway promotes the translation of various proteins, including α -syn. Rapamycin, however, can reduce α -syn translational efficiency by suppressing mTOR activity (Khan et al., 2023). In PD mouse models, rapamycin's therapeutic effect is notable: striatal α -syn levels are decreased, DA release is elevated, and dopaminergic

synaptic signal transmission function is markedly improved. This provides a novel therapeutic approach for alleviating motor dysfunction in PD patients (Liu et al., 2013).

6.1.2. Inhibiting pathogenic protein aggregation

In the field of anti-A β aggregation agents, the monoclonal antibody solanezumab is a prominent candidate. It can selectively engage with soluble A β O, not only directly inhibiting A β O's further aggregation but also blocking synaptic damage induced by A β O binding (Doody et al., 2014). This effect has been clearly validated in clinical trials involving AD patients: post-solanezumab treatment, CSF A β O levels are significantly reduced, and concurrently, the decline rate of hippocampal synaptic density is slowed—offering potential for delaying AD-related cognitive decline. Beyond monoclonal antibodies, natural compounds also possess distinctive merits in anti-A β aggregation research, with curcumin being widely investigated (Doody et al., 2014). Curcumin can mitigate pathological aggregation by inhibiting A β fibrillation. In the APP/PS1 transgenic mouse model, curcumin treatment reduces the number of hippocampal A β plaques and enhances synaptic transmission efficacy. Compared with some chemically synthesized drugs, curcumin exhibits higher safety, providing a new option for AD adjuvant therapy (Zhang et al., 2023).

Similar to AD's pathogenic mechanism, PD onset is closely linked to the aberrant aggregation and toxic spread of α -synO. Thus, anti- α -syn aggregation agents have become a key direction in PD therapeutic research. The monoclonal antibody prasinezumab can selectively bind to α -synO, effectively preventing its prion-like spread in the brain and thereby reducing damage to dopaminergic neurons and synapses (Pagano et al., 2022). In PD mouse model experiments, prasinezumab treatment reduces substantia nigra α -synO levels, decreases dopaminergic synaptic loss rate, and significantly improves the model mice's motor dysfunction (Pagano et al., 2024). In addition to antibody-based therapies, small-molecule compounds have also advanced in anti- α -syn aggregation research. For example, anle138b can directly inhibit α -syn aggregation. In the A53T transgenic mouse model, anle138b treatment reduces striatal α -syn aggregation and markedly improves DA synaptic function. This finding provides novel targets and ideas for the development of PD small-molecule drugs (Wagner et al., 2013).

Studies on the aforementioned two classes of agents demonstrate that targeted intervention against pathological oligomers in neurodegenerative diseases can delay disease progression by reducing pathogenic protein aggregation and preserving synaptic function. This offers a more precise therapeutic approach for diseases such as AD and PD. With the advancement of clinical research, such agents may deliver greater therapeutic value to patients in the future.

6.1.3. Promoting pathogenic protein degradation

The autophagic system, acting as the “garbage disposal station” within cells, can effectively accelerate the degradation of pathogenic proteins when its function is enhanced. Rapamycin, an mTOR inhibitor, is a classic tool for activating autophagy (Civiletto et al., 2018). This intervention method has demonstrated synaptic protection effects in various disease models. In AD mice, treatment with rapamycin increases the degradation of A β in the hippocampus, significantly alleviating the previously impaired synaptic structure and function (Cai et al., 2024). In PD mice, rapamycin also exerts its effect: it not only increases the degradation of α -syn in the substantia nigra but also promotes an increase in dopaminergic synaptic density. Meanwhile, it improves the homeostatic balance of synapse-related proteins, achieving multi-dimensional protection of synapses (Liu et al., 2023).

Besides the autophagic system, the UPS is also a key pathway for clearing abnormal proteins, and it plays an important role, especially in the degradation of p-tau and mHTT. Bortezomib, a proteasome activator, can strengthen this process by enhancing proteasome activity. In AD mice, treatment with bortezomib reduces the concentration of p-tau in the hippocampus, improving abnormalities in the substance transport

function of the presynaptic membrane (Vulin et al., 2023); In HD mice, bortezomib can increase the degradation of mHTT in the striatum, restoring impaired synaptic plasticity (Dasgupta et al., 2015). Since bortezomib can act on characteristic pathogenic proteins of different diseases, this also provides the possibility for developing universal therapeutic regimens for multiple neurodegenerative diseases.

6.2. Regulating synaptic protein homeostasis

6.2.1. Promoting synaptic protein synthesis

Activating the CREB pathway upregulates transcriptional levels of synapse-associated proteins via gene expression modulation. Forskolin, in turn, promotes CREB phosphorylation by activating the cAMP/PKA pathway, thereby initiating this regulatory cascade. This strategy has demonstrated synaptic repair efficacy across diverse disease models (Kapoor et al., 2022). In AD mice, forskolin treatment elevates hippocampal CREB phosphorylation, increases downstream BDNF expression, and ultimately boosts synaptic density (Rosa and Fahnstock, 2015); In PD mice, forskolin also exerts therapeutic effects: it enhances striatal CREB activity and DA receptor expression, improves postsynaptic membrane signal transduction efficacy, and mitigates dopaminergic synaptic dysfunction (Alharbi et al., 2022).

Beyond transcriptional regulation, promoting local synaptic translation directly and rapidly supplements synapse-required proteins, addressing the immediate demands of synaptic function repair. 4E-BP1 inhibitors facilitate synaptic mRNA translation by activating eukaryotic translation initiation factor 4E (eIF4E), a strategy that has shown substantial therapeutic benefits in specific diseases. In HD mice, 4E-BP1 inhibitors increase striatal glutamate transporter VGLUT1 mRNA translation, elevate glutamate release, and improve excitatory synaptic transmission (Talukder et al., 2021); In ALS mice, these inhibitors promote motor neuron AChR mRNA translation, repair NMJ function, and further alleviate murine muscle weakness (Peng et al., 2022).

6.2.2. Improving synaptic protein folding

In synaptic protection interventions for neurodegenerative diseases, misfolding and aberrant aggregation of synaptic proteins are major drivers of synaptic structural impairment and functional loss. Thus, maintaining synaptic proteins in their correctly folded conformation has become a pivotal direction for reducing pathological damage and preserving synaptic function.

Geldanamycin is a widely used molecular chaperone activator. It selectively activates heat shock protein 70 (HSP70), whose chaperone function promotes the correct folding of synapse-associated proteins (Zatsepina et al., 2021). This intervention strategy has demonstrated clear synaptic protection effects in various neurodegenerative disease models. In AD mice, treatment with geldanamycin increases the expression level of HSP70 in the hippocampus, and the folding efficiency of PSD-95 (a postsynaptic membrane scaffold protein regulated by HSP70) increases. This further promotes the normal aggregation of postsynaptic membrane receptors and reduces postsynaptic membrane structural abnormalities caused by protein misfolding (Zare et al., 2015); In PD mice, geldanamycin also acts effectively: it boosts substantia nigra HSP70 activity, increases the proportion of correctly folded α -syn, effectively mitigates neurotoxicity from aberrant α -syn aggregation, and alleviates dopaminergic synaptic impairment (Shen et al., 2005).

6.2.3. Enhancing synaptic protein degradation

In synaptic protection interventions for neurodegenerative diseases, clearing abnormal synaptic proteins generated by pathological factors is a critical step to restore synaptic structure and function. Autophagy modulators achieve this by activating the autophagic system. Among these, trehalose operates independently of the classical mTOR pathway, regulating autophagic activity more gently and providing an effective route for clearing abnormal synaptic proteins in diverse diseases.

This mTOR-independent autophagy activation has demonstrated targeted synaptic repair effects in both AD and HD models. In AD mice, trehalose treatment significantly elevates hippocampal autophagic levels, enhancing degradation of abnormally accumulated presynaptic protein SYN, markedly improving previously impaired presynaptic membrane structure, and aiding in restoring presynaptic neurotransmitter release function (Portbury et al., 2017); In HD mice, trehalose targets the striatum, promoting degradation of abnormally aggregated postsynaptic scaffold protein Shank3, gradually restoring damaged PSD structure, further improving synaptic plasticity, and alleviating HD-related synaptic transmission disorders (Tanaka et al., 2004).

6.3. Improving excitotoxicity and calcium homeostasis

6.3.1. Inhibiting excessive activation of NMDAR

In neurodegenerative diseases, calcium signaling dysregulation induced by abnormal NMDAR function is a key mechanism underlying synaptic impairment. Addressing this requires regulating NMDAR activity to balance calcium signaling and protect synaptic function—an intervention direction divided into two specific strategies: selective blocking of toxic signals, and moderate regulation of activity balance, corresponding to the use of GluN2B subunit-selective antagonists and NMDAR partial agonists, respectively.

GluN2B subunit-selective antagonists precisely target GluN2B-containing NMDARs, specifically blocking toxic Ca^{2+} influx mediated by these receptors and preventing synaptic impairment from calcium overload (Wu et al., 2017). Ifenprodil and Ro25-6981 are typical representatives of such antagonists, both demonstrating synaptic protection effects across various disease models. In AD transgenic mice, Ro25-6981 treatment reduces hippocampal NMDAR-mediated Ca^{2+} influx, alleviates suppressed LTP, increases postsynaptic scaffold protein PSD-95 expression, and promotes postsynaptic structural stability (Li et al., 2011); In ALS mice, ifenprodil decreases NMDAR activity in spinal anterior horn motor neurons, reduces Ca^{2+} overload, and ultimately increases NMJ synaptic density, alleviating motor synaptic impairment (Mirmotahari et al., 2024). In contrast to the “blocking” effect of antagonists, NMDAR partial agonists maintain normal synaptic plasticity while avoiding toxicity from excessive receptor activation by moderately activating NMDARs, achieving functional balance. D-cycloserine is a representative agonist that has shown significant efficacy in PD mouse models. It precisely regulates NMDAR activity in MSNs, improves the glutamate-dopamine signaling imbalance induced by dopaminergic synaptic impairment, increases synaptic transmission efficiency, and reduces murine motor symptom severity. This drug also exhibits good safety during action, providing a mild and effective solution for synaptic function repair in PD (Rouaud and Billard, 2003).

6.3.2. Regulating calcium channel activity

In interventions targeting cell membrane and intracellular calcium channels, the repurposing of established drugs and development of novel modulators synergize to deliver precision solutions for calcium dysregulation in diverse neurodegenerative diseases. Repurposing nimodipine represents a breakthrough in AD treatment. It selectively inhibits $\text{CaV}1.2$ channels in pericytes, alleviates capillary constriction, restores cerebral blood flow, and reverses $\text{A}\beta$ -induced pericyte contraction in human cortical tissues (Korte et al., 2024). The novel felodipine liposomal nanodrug (felodipine@LND) crosses the BBB with assistance from low-intensity pulsed ultrasound, inhibits the NLRP3 inflammasome in microglia, reduces $\text{A}\beta$ production, promotes mitophagy, and markedly enhances cognitive performance in AD model mice (He et al., 2024). Additionally, oxcarbazepine diminishes aberrant dopamine release in PD models, ameliorating motor symptoms and improving synaptic plasticity.

Precision optimization of NMDAR antagonists is a recent research focus. Memantine modulates the channel's allosteric site, inhibiting excessive calcium influx while preserving physiological synaptic

transmission, and has entered clinical use for AD (Winblad et al., 2010). TRPV1 modulators such as astaxanthin upregulate RyR2 expression to enhance calcium clearance and inhibit the NF- κ B pathway to reduce synaptic inflammation, exerting synaptic protective effects in both AD and PD (Wang et al., 2023b). For inositol 1,4,5-trisphosphate receptor (IP3R) overactivation in HD, the S1R agonist PRE084 stabilizes the structure of mitochondria-associated endoplasmic reticulum membranes (MERCs), reduces IP3R-mediated calcium release, and protects striatal synapses in HD animals (Hyrskyluoto et al., 2013). Dantrolene suppresses uncontrolled channel opening, restores calcium homeostasis in cerebellar Purkinje cells of spinocerebellar ataxia type 2 (SCA2) model mice, and rescues motor coordination and synaptic plasticity (Liu et al., 2009). Furthermore, flunarizine simultaneously inhibits IP3R and L-type VGCC, significantly improving choreiform movements and cognitive impairment in HD clinical studies (Belfiore et al., 1998). In mitochondrial calcium channel intervention, Ru360 alleviates A β -induced neuronal calcium overload in AD, reduces synaptic loss, and enhances memory (Xie et al., 2017). Gene therapy also shows promise: AFG3L2 gene editing corrects EMRE protein degradation defects in SCA28 and restores MCU complex function; activating the PINK1/-PARKIN pathway repairs the mitochondrial quality control system in PD, thereby indirectly improving synaptic energy supply (Tulli et al., 2019).

6.3.3. Enhancing glutamate clearance

In neurodegenerative diseases, excitotoxicity driven by aberrant glutamate accumulation in the synaptic cleft is a pivotal trigger for synaptic impairment. Addressing this requires reducing synaptic cleft glutamate concentrations through two core intervention strategies: promoting glutamate uptake and accelerating glutamate metabolism. These approaches regulate transporter activity and degrading enzyme function, respectively, forming a synergistic protective mechanism.

Glutamate transporter activators primarily function by enhancing astrocytic glutamate uptake from the synaptic cleft, thereby lowering glutamate levels. Ceftriaxone activates the NF- κ B pathway to upregulate astrocytic glutamate transporter GLT-1 expression and exhibits efficacy across diverse disease models (Rao et al., 2015). In ALS mice, ceftriaxone increases spinal astrocytic GLT-1 expression and improves synaptic cleft glutamate clearance, alleviating motor neuron synaptic impairment and extending murine survival (Rothstein et al., 2005); in AD mice, ceftriaxone reduces hippocampal glutamate concentrations, diminishes excessive NMDAR activation, and ultimately promotes dendritic spine density elevation, preserving synaptic structural integrity (He et al., 2023).

Beyond promoting uptake, glutamate-degrading enzyme enhancers accelerate intracellular and synaptic cleft glutamate catabolism by increasing degrading enzyme activity. NADH effectively facilitates glutamate degradation and has shown significant effects in HD mouse models: NADH treatment boosts striatal glutamate dehydrogenase activity and enhances glutamate catabolism. This not only mitigates excessive postsynaptic NMDAR activation but also improves abnormally augmented LTD, aiding in restoring impaired synaptic plasticity and providing a novel pathway for HD synaptic function repair (Yaku and Nakagawa, 2023).

6.4. Regulating inflammatory response

6.4.1. Inhibiting excessive activation of microglia

In neurodegenerative disease-related synaptic impairment, aberrant microglial activation and inflammatory responses are key driving factors. Overactivated microglia not only release substantial pro-inflammatory factors to damage synapses but also disrupt synaptic structures via abnormal phagocytosis. Thus, regulating microglial activation states and functional phenotypes to reduce neuroinflammation has become a critical intervention direction for synaptic protection.

Microglial activation inhibitors reduce inflammatory synaptic

damage by suppressing abnormal microglial activation and decreasing pro-inflammatory factor release. Minocycline inhibits microglial activation by blocking the NF- κ B pathway and has demonstrated synaptic protective effects across various disease models. In AD mice, minocycline treatment reduces the number of activated hippocampal microglia, decreases pro-inflammatory factor TNF- α and IL-1 β expression, mitigates abnormal synaptic phagocytosis by microglia, and increases pre-synaptic protein SYN expression, effectively preserving presynaptic structures (Biscaro et al., 2012); In PD mice, minocycline inhibits substantia nigra microglial activation, reduces dopaminergic synaptic loss, and increases striatal DA content, alleviating dopaminergic synaptic dysfunction (Wu et al., 2002). In contrast to the “inhibitory” effect of activation inhibitors, microglial phenotype modulators exert synaptic protection by inducing microglial transformation from the pro-inflammatory M1 phenotype to the anti-inflammatory M2 phenotype. IL-4 and IL-13 are typical phenotype modulators that selectively induce this phenotypic shift. In ALS mice, IL-4 injection increases spinal microglial M2 phenotype marker expression, such as Arg1 and CD206, reduces pro-inflammatory factor release, alleviates motor synaptic impairment, reduces muscle atrophy, and ultimately improves murine motor function—providing a novel approach for ALS synaptic protection and symptom alleviation (Chiu et al., 2008).

6.4.2. Blocking complement system activation

In neurodegenerative diseases, aberrant complement system activation damages synapses via a “mark-phagocytosis” mechanism, complement components first tag damaged or excess synapses, and then mediate synaptic phagocytic clearance by immune cells. Targeting this pathological process, blocking the complement cascade by inhibiting key components has emerged as a critical synaptic protection strategy.

As the initiating component of the complement activation pathway, C1q is a pivotal molecule for synaptic tagging. C1q inhibitors directly block synaptic tagging by C1q, thereby reducing subsequent phagocytic damage. C1q-neutralizing antibodies interfere with C1q's synaptic binding capacity and have demonstrated significant efficacy across diverse disease models (Zhong et al., 2023). In AD mice, C1q-neutralizing antibodies reduce hippocampal C1q-synapse binding, decrease the proportion of microglia-phagocytosed synapses, and correspondingly elevate synaptic density (Dejanovic et al., 2018); In HD mice, inhibiting C1q activity mitigates striatal synaptic loss, restores impaired LTP, and further enhances murine cognitive function (Wilton et al., 2023). Complement C3 is a core intermediate component of the complement cascade, and its activation is a key step for the system to exert pathological effects. C3 inhibitors interrupt downstream cascade reactions by blocking C3 activation (Nester et al., 2025). C3 convertase inhibitors directly suppress C3 activation and have shown targeted protective effects in ALS mice: these inhibitors reduce spinal cord C3 deposition, diminish abnormal activation of CR3 on motor neuron postsynaptic membranes, enhance NMJ synaptic transmission efficacy, effectively alleviate nerve-muscle signal transmission disorders, and provide a novel direction for ALS synaptic function repair (DeLoach et al., 2015).

6.5. Supplementing neurotrophic factors

6.5.1. Exogenous neurotrophic factor supplementation

In neurotrophic factor-mediated synaptic protection, the BBB represents a key barrier limiting therapeutic efficacy. Thus, achieving precise targeted delivery of neurotrophic factors has become a core prerequisite for improving treatment outcomes. Among these factors, BDNF and GDNF have emerged as pivotal research targets due to their critical roles in synaptic repair. Through diverse delivery approaches, both have demonstrated significant efficacy in corresponding neurodegenerative diseases.

As a key factor sustaining synaptic survival and function, BDNF targeted delivery requires BBB penetration. Currently, intracerebral

injection or viral vectors are commonly used to achieve local high expression, a delivery strategy that has shown clear effects in disease models (Xing et al., 2016). In AD mice, hippocampal injection of AAV-BDNF increases regional BDNF expression, which in turn upregulates presynaptic SYN and postsynaptic PSD-95 expression, significantly enhancing murine learning and memory abilities (Jiao et al., 2016); In PD mice, substantia nigra injection of AAV-BDNF increases dopaminergic synaptic density and improves striatal DA release, effectively alleviating PD-related motor symptoms (Chang and Wang, 2021). In contrast to BDNF's broad-spectrum synaptic protective effects, GDNF exerts selective protection on dopaminergic synapses. Its targeted delivery to key PD brain regions significantly improves treatment precision. Striatal injection of AAV-GDNF allows direct GDNF action on dopaminergic synapses, a strategy that has shown excellent performance in PD Phase I/II clinical trials: patients exhibit increased striatal DAT density, improved dopaminergic synaptic function, elevated motor scores, and favorable safety profiles throughout treatment. This provides a reliable targeted therapy for PD synaptic function repair and clinical symptom alleviation (Van Laar et al., 2025).

6.5.2. Regulating endogenous neurotrophic factors

Intervention strategies for BDNF-mediated synaptic protection fall into two core categories: "promoting endogenous BDNF production" and "mimicking BDNF physiological functions." The former enhances endogenous BDNF levels by activating BDNF transcriptional processes, while the latter bypasses BDNF delivery challenges by directly activating BDNF receptors. These approaches act on distinct pathways to synergistically protect synapses.

BDNF transcription activators promote endogenous BDNF production and expression by regulating transcription-related signaling pathways. Forskolin is a typical representative, achieving this via CREB pathway activation (Kapoor et al., 2022). In HD mice, forskolin treatment increases striatal CREB phosphorylation, boosts BDNF mRNA transcription, and further induces upregulation of postsynaptic signaling molecule DARPP-32, restoring impaired synaptic plasticity (Obrietan and Hoyt, 2004). Beyond regulating endogenous BDNF production, direct activation of BDNF's specific receptor (TrkB) to mimic BDNF physiological functions is another important synaptic protection. TrkB agonists activate downstream receptor pathways independently of endogenous BDNF expression (Gao et al., 2016). 7,8-Dihydroxyflavone has demonstrated synaptic protective effects across various disease models. In AD mice, this agent increases hippocampal TrkB phosphorylation, activates the PI3K/Akt signaling pathway, promotes synaptic protein synthesis, and restores impaired LTP (Gao et al., 2016); In PD mice, 7,8-Dihydroxyflavone activates TrkB receptors in substantia nigra dopaminergic neurons, mitigates α -synuclein (α -syn)-induced synaptic damage, and preserves the integrity of dopaminergic neural circuits (Nie et al., 2019).

6.6. Mitophagy enhancer-mediated synaptic protections

As a potential therapeutic strategy for synaptic dysfunction, mitophagy enhancers can synergistically reverse synaptic damage through multiple dimensions including clearing toxic substrates, restoring energy supply, and repairing structural integrity by targeting and regulating mitochondrial quality control pathways. Their mechanism of action is supported by clear experimental evidence and holds promising potential for clinical translation (Morton et al., 2021).

These enhancers can accelerate the selective clearance of damaged mitochondria in synaptic regions by activating the PINK1/Parkin pathway and regulating the expression of autophagy receptors (Pradeepkiran and Reddy, 2020). In cellular models of AD, molecules such as Urolithin A and nicotinamide riboside can significantly upregulate the phosphorylation level of PINK1, promote the recruitment of Parkin to fragmented mitochondria, and significantly increase the clearance efficiency of damaged mitochondria. Meanwhile, they can

activate the adenosine 5'-monophosphate (AMP)-activated protein kinase (AMPK) signaling pathway by inhibiting the activity of mTOR kinase, relieve the autophagy-suppressed state, reduce the accumulation of p62 protein, and prevent synaptic protein oxidative damage induced by ROS released from impaired mitochondria (Kshirsagar et al., 2022a, 2022b; Morton et al., 2021; Pradeepkiran and Reddy, 2020). In HD models, these enhancers can also upregulate the expression of autophagy receptors such as BNIP3 and NIX, enhance the recognition of toxic mitochondria bound to mHTT, reduce the level of oxidative stress in synaptic regions, and alleviate neuronal excitotoxicity (Sawant et al., 2021). Beyond clearing damaged mitochondria, mitophagy enhancers can reestablish synaptic energy homeostasis through bidirectional "clearance-biogenesis" regulation. Experiments have shown that in HT22 hippocampal neurons treated with Urolithin A, activation of mitophagy restores the activity of respiratory chain complex I, increases oxidative phosphorylation (OXPHOS) efficiency, and significantly raises synaptic ATP production compared with the model group, thus significantly improving the energy foundation required for neurotransmitter release. The key mechanism lies in the ability of these enhancers to activate peroxisome proliferator-activated receptor gamma coactivator 1-alpha (PGC-1 α) via sirtuin 1 (SIRT1)-mediated deacetylation. This transcriptional coactivator not only promotes mitochondrial biogenesis and increases the density of functionally normal mitochondria in synaptic regions, but also upregulates Mitofusin2 (Mfn2) expression, restores the balance of mitochondrial fusion/fission, and further optimizes energy metabolism efficiency (Kshirsagar et al., 2022b; Liu et al., 2022). In PD models, nicotinamide riboside activates mitophagy by increasing NAD⁺ levels, which elevates ATP production at the synapses of dopaminergic neurons in the substantia nigra and reverses neurotransmitter release disorders caused by energy deficiency. Mitophagy enhancers can also directly protect synaptic architecture by reducing toxic accumulation (Schöndorf et al., 2018). In AD mouse models, after Urolithin A intervention, the expression levels of synaptophysin and PSD95 are significantly increased compared with the control group, the thickness of the postsynaptic density is restored to normal, the synaptic connection density in the hippocampus is notably elevated, and these changes are accompanied by improved spatial memory ability. This effect is closely related to the enhancers' ability to reduce the formation of A β -Drp1 complexes and inhibit excessive tau phosphorylation; by clearing damaged mitochondria, they cut off the amplification loop of toxic molecules and prevent synaptic protein degradation (Hou et al., 2024; Pradeepkiran et al., 2022). In HD models, PGC-1 α agonists enhance mitophagy, which improves the synaptic transmission efficiency of medium spiny neurons in the striatum and significantly enhances motor coordination, confirming their role in repairing synaptic transmission function (Tsunemi et al., 2012).

Notably, different types of mitophagy enhancers exhibit specific effects (Kshirsagar et al., 2025): Urolithin A has the optimal efficacy in restoring mitochondrial dynamic balance; nicotinamide riboside is more suitable for synaptic disorders associated with energy metabolism imbalance and PGC-1 α agonists show clear synaptic protective activity in various neurodegenerative disease models. Studies indicate that the combined use of mitophagy enhancers and Drp1 inhibitors produces a synergistic effect. This provides an important clinical insight for the multi-target combination therapy of neurodegenerative diseases and further highlights the core value of mitophagy regulation in synaptic function protection.

6.7. Other novel therapeutic strategies

6.7.1. Stem cell transplantation

In neurodegenerative disease synaptic protection, mesenchymal stem cells (MSCs) have become a key tool for regulating the neural microenvironment and repairing damaged synapses, leveraging their unique paracrine properties. Unlike direct neurotransmitter supplementation or single signaling pathway regulation, MSCs exert multi-

dimensional effects—including anti-inflammation and nutritional support—by secreting various bioactive factors, creating favorable conditions for synaptic repair (Seo and Cho, 2012).

MSC-mediated synaptic protection primarily relies on secreted factors such as BDNF and IL-10. These factors not only suppress excessive inflammatory responses but also provide nutritional support for synaptic repair, a characteristic validated across diverse disease models (Wang et al., 2013). In AD mice, intravenous MSC injection increases hippocampal BDNF and IL-10 expression, effectively reducing synaptic cleft glutamate concentrations to avoid excitotoxic damage, increasing synaptic density, and enhancing synaptic structural integrity (Guo et al., 2020; Terashima et al., 2020).

6.7.2. Regulating gut microbiota

In neurodegenerative disease synaptic protection research, the gut-brain axis serves as a key pathway connecting the gut and CNS, offering a novel peripheral intervention perspective for synaptic function regulation. Probiotic intervention achieves synaptic protection by modulating gut-brain axis function and influencing the CNS microenvironment via gut microbiota metabolites. This approach is non-invasive, easy to implement, and particularly suitable for long-term intervention (Yang et al., 2023).

The core of probiotic action lies in improving gut microbiota composition and promoting the production of beneficial metabolites such as short-chain fatty acids. These metabolites act on the CNS via the circulatory system or neural pathways, regulating inflammatory responses and neurotrophic factor levels to indirectly protect synaptic structure and function (Zhao et al., 2024). This mechanism has been validated in various disease models. In AD mice, *Bifidobacterium* supplementation increases gut short-chain fatty acid content, such as butyrate. This metabolite inhibits central inflammation via the gut-brain axis, reducing hippocampal microglial activation, and promotes BDNF upregulation, effectively alleviating synaptic damage (Zhu et al., 2021); In PD mice, *Lactobacillus* supplementation modulates α -syn abnormal aggregation in the substantia nigra via the gut-brain axis, increases dopaminergic synaptic density, and improves motor symptoms. Furthermore, this non-invasive intervention is easy to adhere to long-term, providing a feasible solution for neurodegenerative disease long-term management (Aktas et al., 2024).

7. Challenges

While our understanding of synaptic dysfunction in age-related neurodegenerative disorders has improved notably, several key challenges remain unresolved. A major obstacle lies in translating preclinical success to human outcomes. Many therapies that work well in experimental models struggle to deliver consistent effects in patients. This is partly due to difficulties penetrating the BBB to target relevant brain regions, and partly because single-target treatments cannot address the overlapping pathological mechanisms driving these diseases. Research into rare neurodegenerative diseases – such as Dementia with DLB and FTD – remains limited. Critical details like the synergistic toxic effects between α -synuclein and A β in DLB are not fully understood. Disease-specific biomarkers are also still absent, leaving these conditions understudied. Newly identified biomarkers require further verification before clinical use. Minimally invasive markers, including synaptic protein fragments in blood and synapse-related proteins in exosomes, need large-cohort studies to validate their diagnostic sensitivity and specificity. These unresolved issues impede the advancement of early diagnostic methods and effective treatments, highlighting the urgency of targeted approaches to close the gap between preclinical research and clinical application.

8. Conclusion and future directions

Despite the aforementioned advances, critical challenges remain

unresolved. Many therapeutic strategies efficacious in preclinical settings have not yielded consistent efficacy in clinical trials, highlighting the need to refine drug delivery systems such as BBB penetration and formulate multi-target combination therapies to disrupt pathological networks. Currently, core challenges manifest in three key areas. First, clinical translation bottlenecks persist. Most interventions effective in animal models such as γ -secretase inhibitors and BDNF supplementation struggle to stably ameliorate patient symptoms in clinical settings, largely due to difficulties in BBB penetration and the synergistic interplay of multiple pathological mechanisms. Second, insufficient research into rare neurodegenerative diseases exists. Mechanisms of synaptic damage such as the interactive toxicity between α -synuclein and A β in DLB, as well as disease-specific biomarkers for conditions like DLB and FTD, remain poorly elucidated, resulting in incomplete research coverage. Third, validation of novel biomarkers is inadequate. Minimally invasive indicators including exosome-encapsulated synapse-related proteins and blood-borne synaptic fragments require validation of their diagnostic sensitivity and specificity in larger cohorts before clinical application.

To address these core challenges, future research can achieve breakthroughs in three dimensions. First, promote multi-target combination therapies. Target mechanistic intersections associated with pathological protein toxicity and excitotoxicity, such as concurrent inhibition of A β aggregation and aberrant NMDAR activation, and integrate these with innovative delivery systems, including nanocarriers and nasal mucosal administration, to enhance CNS-targeting efficacy. Second, expand research into rare neurodegenerative diseases. Utilize iPSC-derived models and high-resolution imaging technologies to clarify the specificity of synaptic damage in DLB and FTD, thereby completing the disease spectrum. Third, deepen mechanistic insights and accelerate biomarker translation. Establish a multi-dimensional combined detection system encompassing CSF, blood, and imaging to promote the clinical translation of biomarkers. Additionally, leverage technologies such as single-cell sequencing and brain organoids to advance the understanding of synaptic damage mechanisms.

In summary, targeted synaptic protection represents a pivotal frontier in the intervention of age-related neurodegenerative diseases. Advancing the understanding of the network mechanisms underlying synaptic damage, refining early diagnostic biomarkers, and developing clinically translatable therapeutic strategies will be the core directions for delaying disease progression, improving patient outcomes, and ultimately addressing the unmet medical needs of these devastating conditions.

Abbreviations

AD, Alzheimer's disease; PD, Parkinson's disease; HD, Huntington's disease; ALS, Amyotrophic lateral sclerosis; DLB, Dementia with Lewy Bodies; FTD, Frontotemporal Dementia; FTLD-TDP, Frontotemporal lobar degeneration with TDP-43 inclusions; A β , Amyloid- β ; A β O, A β oligomers; p-tau, Hyperphosphorylated tau; NFTs, Neurofibrillary tangles; α -syn, α -synuclein; α -synO, α -synuclein oligomers; LBs, Lewy bodies; LNs, Lewy neurites; mHTT, Mutant huntingtin; TDP-43, TAR DNA-binding protein 43; SOD1, Superoxide dismutase 1; FUS, Fused in sarcoma; PrPC, Prion protein; CNS, Central nervous system; NMJ, Neuromuscular junctions; MSN, Medium spiny neuron; iPSCs, Induced pluripotent stem cells; PSD, Postsynaptic density; MCU, Mitochondrial calcium uniporter; MERCs, Mitochondria-associated endoplasmic reticulum membranes; BBB, Blood-brain barrier; MSC, Mesenchymal stem cells; NMDAR, N-methyl-D-aspartate receptor; AMPAR, α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor; DA, Dopamine; DAT, Dopamine transporter; SNARE, Soluble N-ethylmaleimide-sensitive factor attachment protein receptor; VAMP2, Vesicle-associated membrane protein 2; IP3R, Inositol 1,4,5-trisphosphate receptors; CaV, Voltage-gated calcium channel; TRPV1, Transient receptor potential vanilloid 1; RyR2, Ryanodine receptor 2; ACh, Acetylcholine; AChR, Acetylcholine receptors; GLT-1, Glutamate transporter 1; EAAT2, Excitatory amino acid transporter 2; VGLUT1, Vesicular glutamate

transporter 1; TrkB, Tyrosine kinase B; GSK-3 β , Glycogen synthase kinase 3 β ; PI3K/Akt, Phosphoinositide 3-kinase/protein kinase B; MAPK/ERK, Mitogen-activated protein kinase/extracellular signal-regulated kinase; PKA, Protein kinase A; CaMKII, Calcium/calmodulin-dependent protein kinase II; CREB, cAMP response element-binding protein; NF- κ B, Nuclear factor- κ B; NLRP3, NOD-like receptor pyrin domain-containing 3; mTOR, Mammalian target of rapamycin; eIF4E, Eukaryotic translation initiation factor 4E; 4E-BP1, Eukaryotic translation initiation factor 4E-binding protein 1; HSP70, Heat shock protein 70; nNOS, Neuronal nitric oxide synthase; TH, Tyrosine hydroxylase; UPS, Ubiquitin-proteasome system; CSF, Cerebrospinal fluid; SYN, Synapsin; PSD-95, Postsynaptic density protein 95; Ng, Neurogranin; NPTX2, Neuronal pentraxin 2; BDNF, Brain-derived neurotrophic factor; NGF, Nerve growth factor; GDNF, Glial cell line-derived neurotrophic factor; MMSE, Mini-Mental State Examination; SYN-F, Synapsin fragments; PSD-F, PSD-95 fragments; MDA, Malondialdehyde; ROS, Reactive oxygen species; DAMPs, Damage-associated molecular patterns; HTT, Huntingtin gene; GRN, Granulin gene; MAPT, Microtubule-associated protein tau gene; C9orf72, Chromosome 9 open reading frame 72 gene; APP, Amyloid precursor protein; PS1, Presenilin 1; siRNA, Small interfering RNA; AAV, Adeno-associated virus.

CRedit authorship contribution statement

X.L., R.W. and C.L. designed and wrote the manuscript. X.L., R.W. and Y.Z. produced figures. Y.W., X.C. and Q.L. edited the manuscript. Q.L. and C.L. designed and edited the manuscript. All authors have read and approved the final version of the manuscript.

Ethics approval

Not applicable.

Informed consent statement

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

No data was used for the research described in the article.

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